Executive Summary — Surrogate Model Validation Report

Use case: UCLOADS_1 | 01 September 2026 | 5 model(s)

REQUIREMENTS PARTIALLY MET

Recommended model: PyTorchNN

Avg R^2: 0.9923 — Excellent

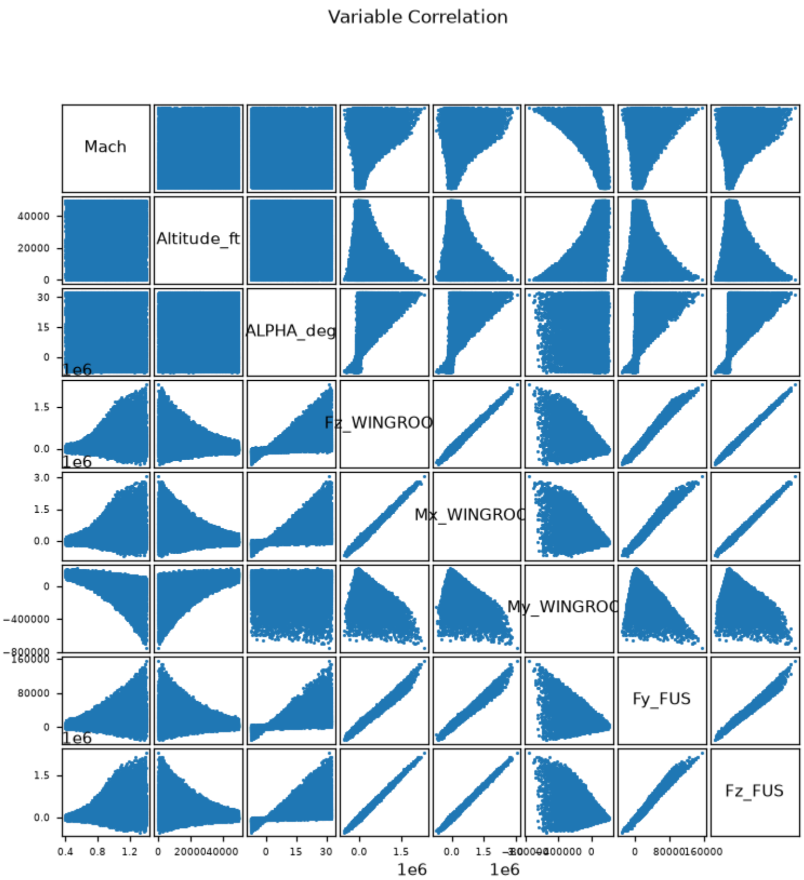
Q90 Accuracy — No output passes

Fail: Fz_WINGROOT, Mx_WINGROOT, My_WINGROOT, Fy_FUS, Fz_FUS, Mx_FUS, My_FUS, Mz_FUS

Project Overview

Use case	UCLOADS_1
Inputs	Mach, Altitude_ft, ALPHA_deg
Outputs	8: Fz_WINGROOT, Mx_WINGROOT, My_WINGROOT, Fy_FUS, Fz_FUS, Mx_FUS, My_FUS, Mz_FUS
Train / Val / Test	70% / 10% / 20%
Accuracy target	Q90 < 10% relative error per output

Variable Correlation — Input vs Output



Model Selection

Model	Avg R^2	Q90 pass (8)	KS pass (8)
MLP	0.9917	0/8	8/8
GradientBoosting	0.9907	0/8	7/8
RandomForest	0.9917	0/8	0/8
XGBoost	0.9590	0/8	7/8
PyTorchNN ⋆	0.9923	0/8	8/8

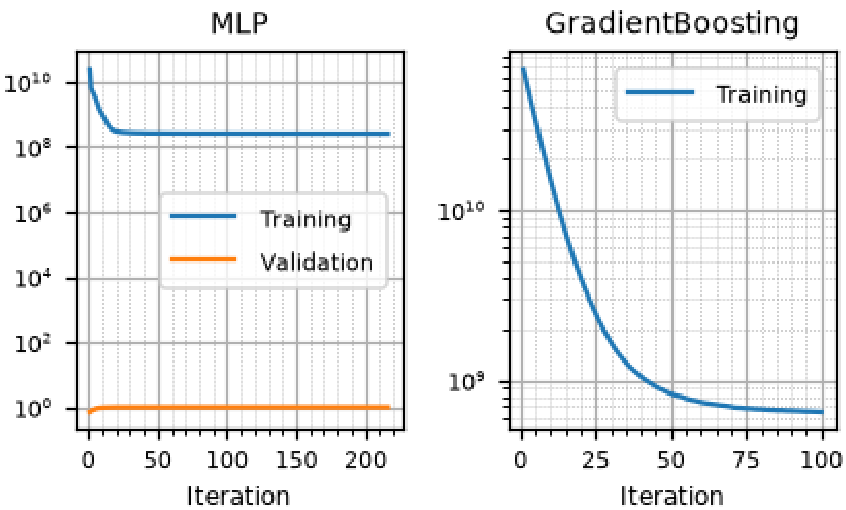
⋆ Recommended model. Q90 target: $< 10\%$. KS: $p \geq 0.05$ = no overfitting.

Accuracy Assessment (Q90 & R^2)

Output	MLP			GradientBoosting			RandomForest			XGBoost			PyTorchNN ⋆		
	R^2	Q90	Gap	R^2	Q90	Gap	R^2	Q90	Gap	R^2	Q90	Gap	R^2	Q90	Gap
Fz_WINGROOT	0.9900	0.2474	+0.1474	0.9880	0.2706	+0.1706	0.9894	0.2555	+0.1555	0.9452	0.5830	+0.4830	0.9901	0.2475	+0.1475
Mx_WINGROOT	0.9900	0.2501	+0.1501	0.9881	0.2733	+0.1733	0.9892	0.2596	+0.1596	0.9452	0.5863	+0.4863	0.9901	0.2494	+0.1494
My_WINGROOT	0.9900	0.2052	+0.1052	0.9896	0.2096	+0.1096	0.9892	0.2131	+0.1131	0.9898	0.2075	+0.1075	0.9901	0.2042	+0.1042
Fy_FUS	0.9876	0.2364	+0.1364	0.9879	0.2344	+0.1344	0.9896	0.2180	+0.1180	0.9462	0.4800	+0.3800	0.9901	0.2133	+0.1133
Fz_FUS	0.9975	0.1262	+0.0262	0.9954	0.1676	+0.0676	0.9972	0.1314	+0.0314	0.9523	0.5560	+0.4560	0.9975	0.1250	+0.0250
Mx_FUS	0.9975	0.1274	+0.0274	0.9954	0.1695	+0.0695	0.9973	0.1319	+0.0319	0.9523	0.5596	+0.4596	0.9975	0.1258	+0.0258
My_FUS	0.9974	0.1038	+0.0038	0.9970	0.1122	+0.0122	0.9969	0.1121	+0.0121	0.9972	0.1075	+0.0075	0.9975	0.1023	+0.0023
Mz_FUS	0.9836	0.2595	+0.1595	0.9840	0.2566	+0.1566	0.9851	0.2485	+0.1485	0.9437	0.4527	+0.3527	0.9859	0.2423	+0.1423

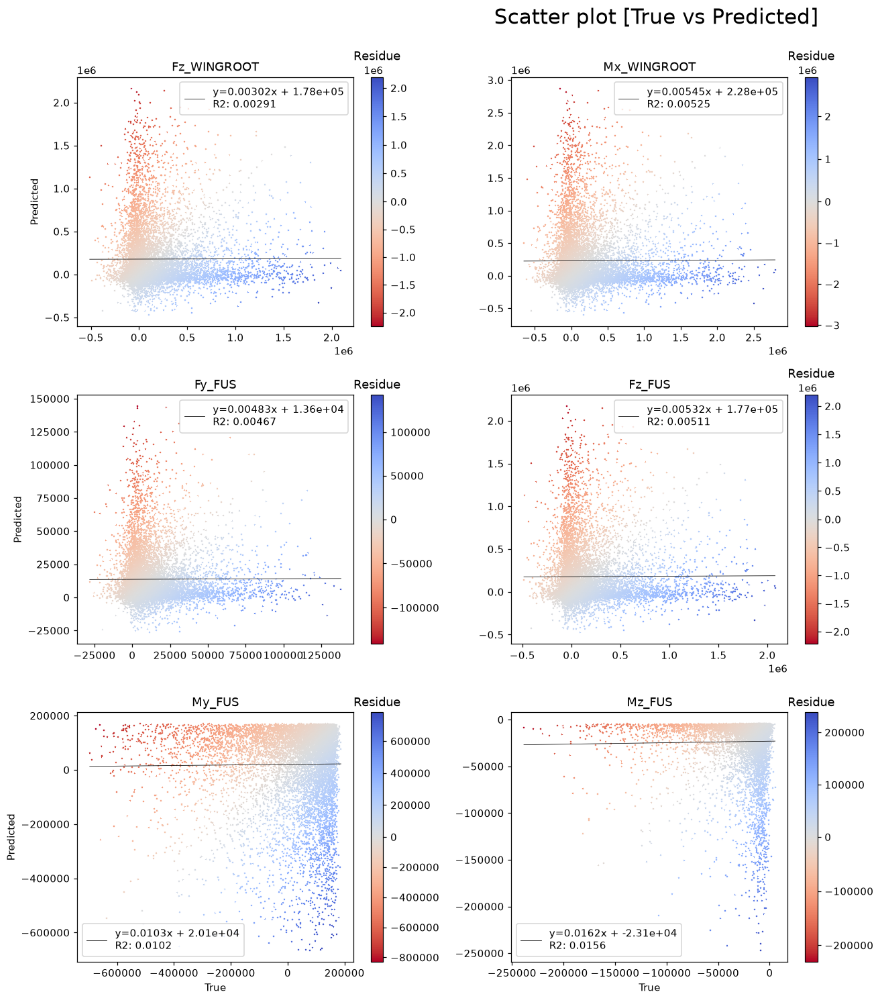
$R^2 \geq 0.95$ $R^2 \in [0.80, 0.95)$ $R^2 < 0.80$ or Q90 fail pass Gap=Q90−target ($<0 \Rightarrow$ margin)

Training History



Training loss per iteration, log scale. A validation panel is shown for models trained with early stopping. Gradient boosting reports per-stage deviance averaged over its per-output estimators.

Predicted vs True — PyTorchNN



Data Split Quality

Metric	Value	Pass
Residual voxel proportion (≤ 0.05)	0.000	✓
Valid test proportion	0.966	
Phacking test proportion	0.005	
Isolated test proportion	0.029	
Chi^2 p-value (≥ 0.05)	—	—

Overfitting Check (KS Test) — PyTorchNN

Output	KS p-value
Fz_WINGROOT	0.891
Mx_WINGROOT	0.172
My_WINGROOT	0.663
Fy_FUS	0.586
Fz_FUS	0.501
Mx_FUS	0.523
My_FUS	0.791
Mz_FUS	0.479

$p \geq 0.05$ — no overfitting $p < 0.05$ — overfitting detected

Deep Analysis

UCLOADS_1

Winner Model — Full Validation

Deep Analysis — PyTorchNN

Mirrors `validation_template.ipynb` — all computations from validation CSVs

Data Overview

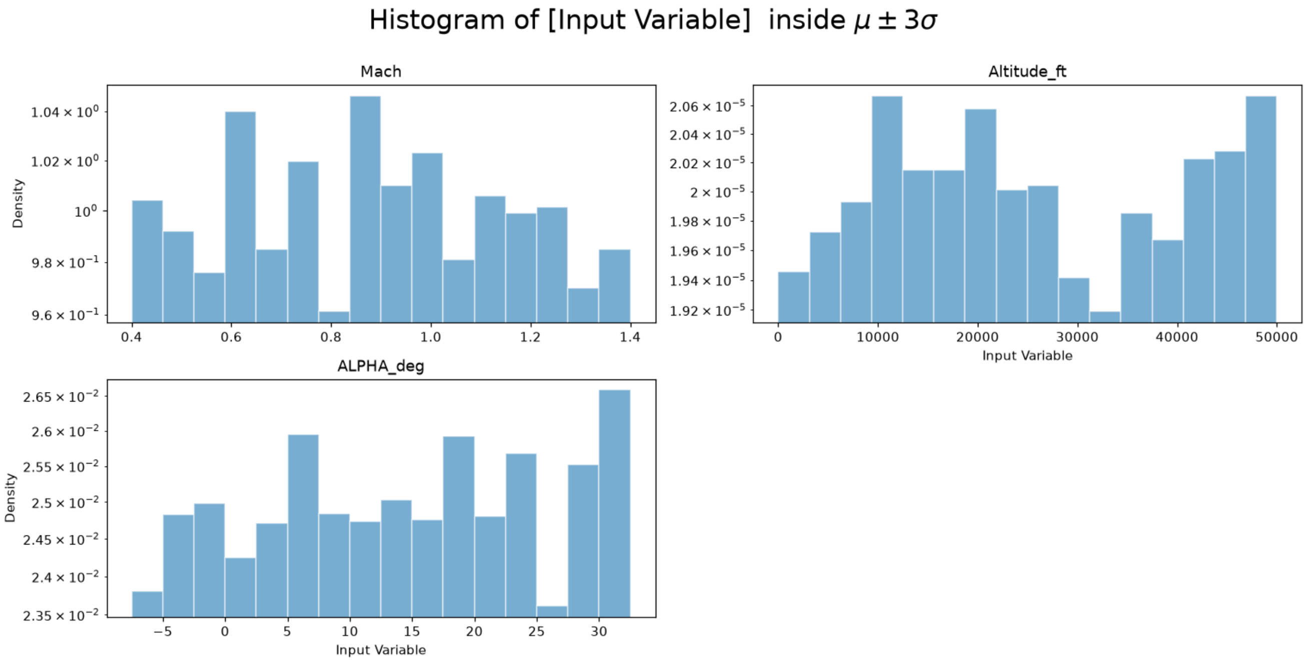
Statistical summary of inputs and outputs over all data (train + val + test), as reported by `describe()` in the feature-selection notebook: count, mean, standard deviation, minimum, quartiles and maximum per variable. All figures in this part are drawn by `validationlib`, matching the extended validation notebook.

Input Statistics

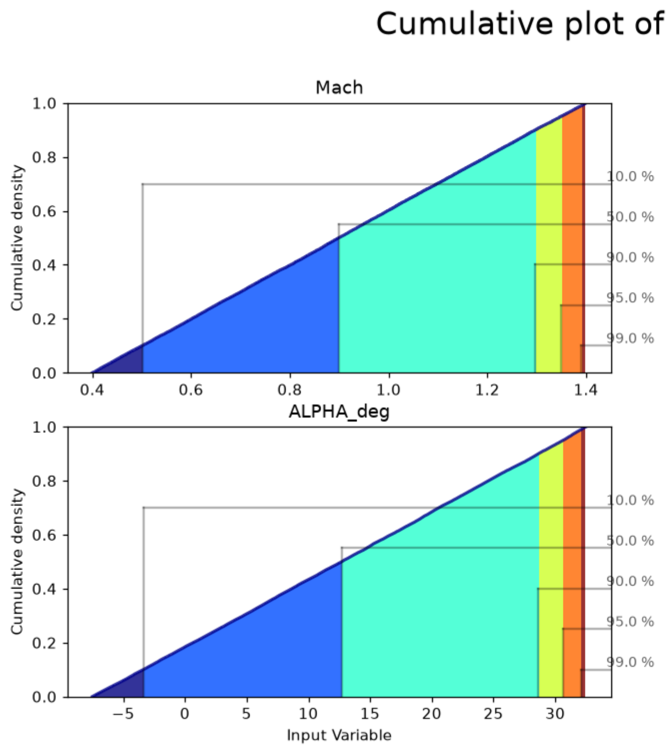
	count	mean	std	min	P25	P50	P75	max
Mach	30,000	0.8993	0.2877	0.4	0.649	0.899	1.148	1.4
Altitude_ft	30,000	2.504e+04	1.445e+04	0.262	1.254e+04	2.491e+04	3.763e+04	5e+04
ALPHA_deg	30,000	12.65	11.54	-7.499	2.714	12.71	22.63	32.5

Input variables — Train_set.describe()

Input Histograms



Input Cumulative Distributions

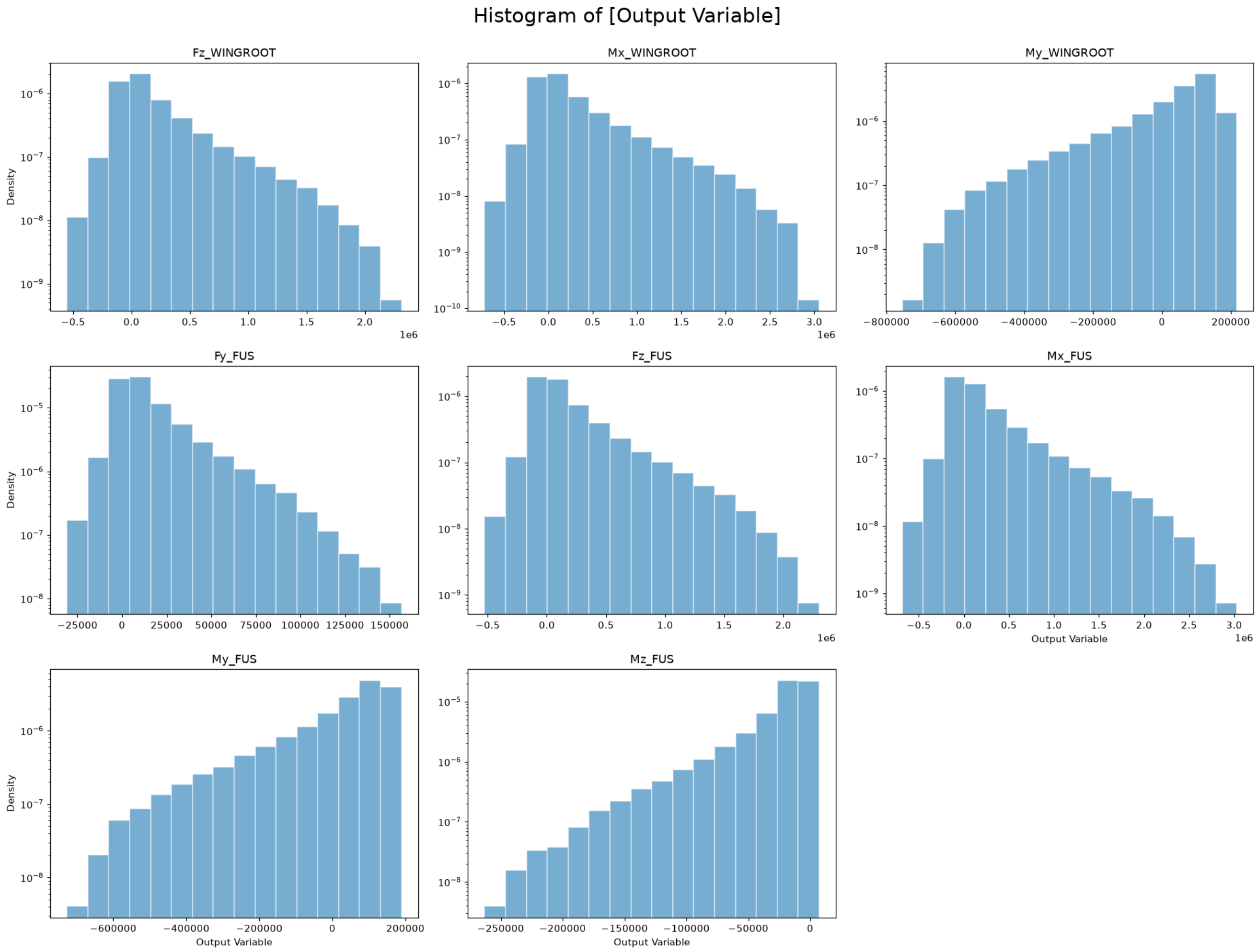


Output Statistics

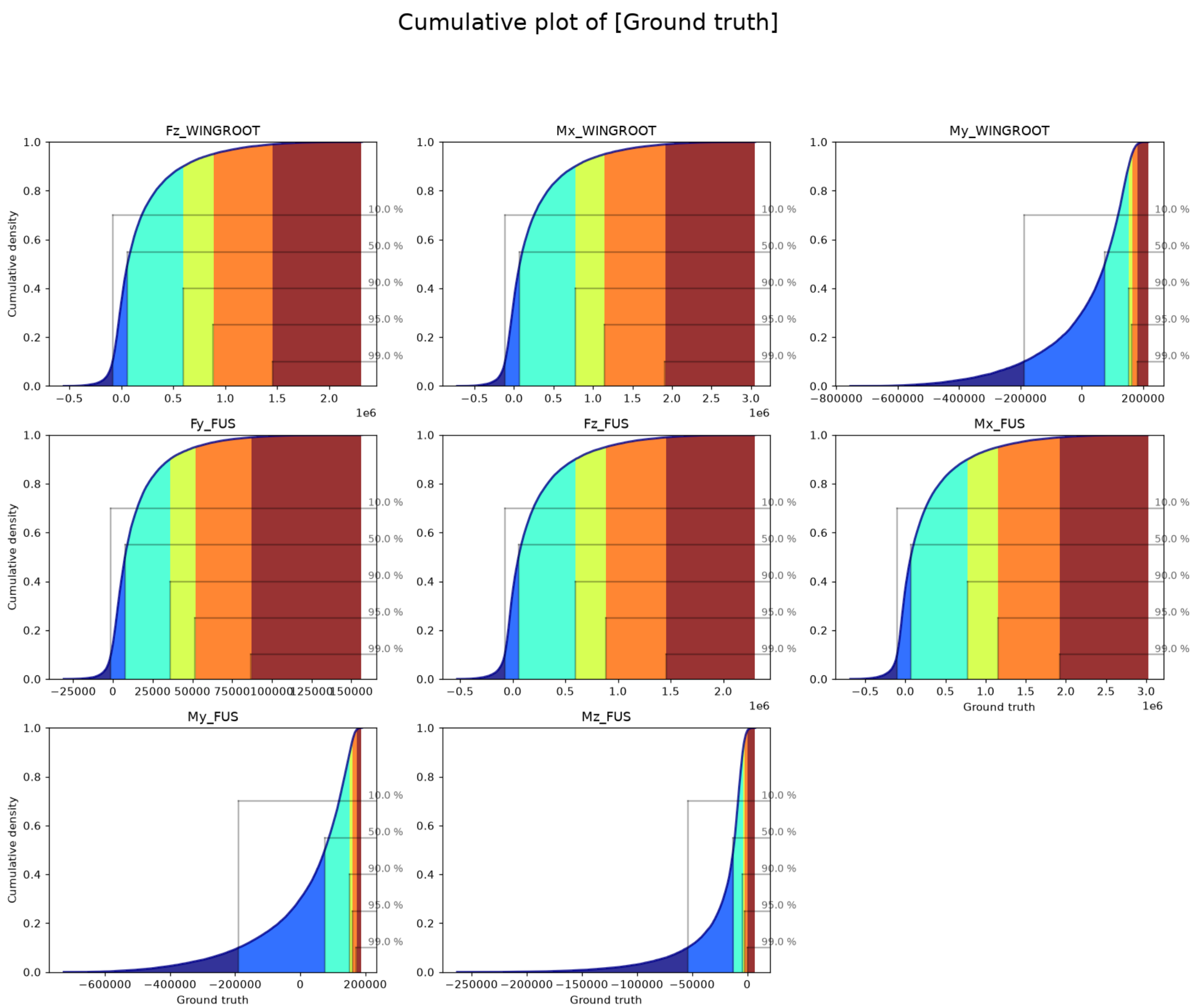
	count	mean	std	min	P25	P50	P75	max
Fz_WINGROOT	30,000	1.707e+05	3.3e+05	-5.524e+05	-2.938e+04	5.923e+04	2.553e+05	2.313e+06
Mx_WINGROOT	30,000	2.198e+05	4.339e+05	-7.196e+05	-4.198e+04	7.218e+04	3.289e+05	3.054e+06
My_WINGROOT	30,000	2.349e+04	1.482e+05	-7.54e+05	-2.958e+04	7.465e+04	1.271e+05	2.17e+05
Fy_FUS	30,000	1.335e+04	1.86e+04	-3.087e+04	2189	7538	1.82e+04	1.566e+05
Fz_FUS	30,000	1.708e+05	3.297e+05	-5.269e+05	-2.881e+04	5.577e+04	2.534e+05	2.303e+06
Mx_FUS	30,000	2.194e+05	4.339e+05	-6.836e+05	-4.365e+04	6.652e+04	3.241e+05	3.028e+06
My_FUS	30,000	2.358e+04	1.482e+05	-7.264e+05	-2.889e+04	7.528e+04	1.276e+05	1.891e+05
Mz_FUS	30,000	-2.266e+04	2.787e+04	-2.63e+05	-2.583e+04	-1.26e+04	-7369	7202

Output variables — Train_set.describe()

Output Histograms



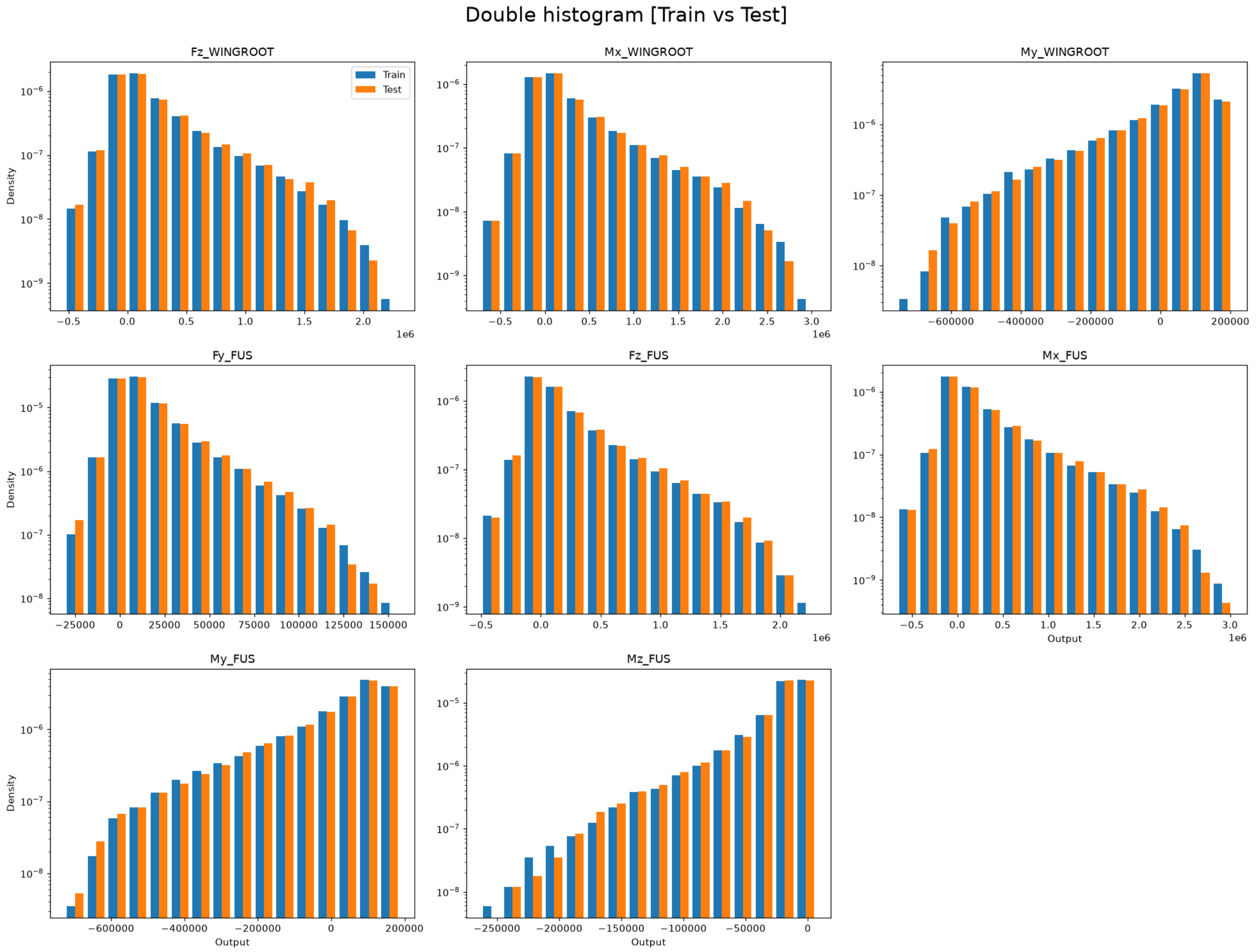
Output Cumulative Distributions


Cumulative plot of [Ground truth]
Fz_WINGROOT
Mx_WINGROOT
My_WINGROOT
Fy_FUS
Fz_FUS
Mx_FUS
My_FUS
Mz_FUS
Cumulative density
Ground truth
1e6
10.0 %
50.0 %
90.0 %
95.0 %
99.0 %

Train-Test Split Analysis

A well-designed split produces training and test distributions that are similar but not identical. The Kolmogorov-Smirnov (KS) test and the Anderson-Darling (AD) test are used to assess distributional similarity. H_0: both sets come from the same distribution. $p < 0.05$ rejects H_0 (undesirable if too low; undesirable if the split was done by stratification and p is too high).

Output Distributions: Train vs Test



Input Distributions: Train vs Test

KS and AD Test Results

	KS	AD
Fz_WINGROOT	0.6637453476707832	0.25
Mx_WINGROOT	0.8629266196033867	0.25
My_WINGROOT	0.7801603722233774	0.25
Fy_FUS	0.8335435086261689	0.25
Fz_FUS	0.8533882178607826	0.25
Mx_FUS	0.9062284518844627	0.25
My_FUS	0.4893600375017084	0.25
Mz_FUS	0.7912059350314521	0.25

Train vs test p-values. Red: $p < 0.05$ (distributions differ).

Error Quantification — P(E)

Two error metrics are studied:

- Residue: $e_i = y_i - \hat{y}_i$. A centred ($\mu \approx 0$), symmetric distribution indicates an unbiased model.
- Absolute Error: $|e_i|$. Always non-negative; its Q90 must satisfy the accuracy requirement.

Residue Statistics Table

Bootstrap confidence bounds shown as superscript (upper) and subscript (lower).

	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
Fz_WINGROOT	10000	-2250000.	2190000.	-4860.	$^{2320.}_{-14000.}$	-1120.	$^{5730.}_{-7150.}$	478000.	$^{487000.}_{469000.}$	395000.	$^{410000.}_{382000.}$	2.53	$^{2.78}_{2.29}$	-0.104	$^{-0.00729}_{-0.196}$
Mx_WINGROOT	10000	-3030000.	2920000.	-7070.	$^{3190.}_{-18800.}$	-885.	$^{5670.}_{-8250.}$	628000.	$^{639000.}_{619000.}$	516000.	$^{530000.}_{496000.}$	2.61	$^{2.84}_{2.37}$	-0.112	$^{-0.0158}_{-0.195}$
My_WINGROOT	10000	-834000.	780000.	2170.	$^{6730.}_{-1990.}$	2370.	$^{5330.}_{233.}$	210000.	$^{213000.}_{207000.}$	201000.	$^{209000.}_{194000.}$	1.39	$^{1.50}_{1.29}$	-0.00989	$^{0.0599}_{-0.0711}$
Fy_FUS	10000	-141000.	143000.	-244.	$^{264.}_{-793.}$	-66.1	$^{304.}_{-442.}$	26800.	$^{27400.}_{26200.}$	22400.	$^{23100.}_{21700.}$	3.17	$^{3.43}_{2.91}$	-0.114	$^{0.0159}_{-0.231}$
Fz_FUS	10000	-2220000.	2200000.	-4950.	$^{5950.}_{-15500.}$	213.	$^{5270.}_{-4580.}$	478000.	$^{492000.}_{470000.}$	395000.	$^{406000.}_{383000.}$	2.55	$^{2.71}_{2.37}$	-0.103	$^{-0.00212}_{-0.199}$
Mx_FUS	10000	-2980000.	2940000.	-6810.	$^{2810.}_{-18000.}$	489.	$^{5200.}_{-6010.}$	629000.	$^{644000.}_{618000.}$	513000.	$^{529000.}_{503000.}$	2.63	$^{2.85}_{2.39}$	-0.105	$^{-0.0110}_{-0.204}$
My_FUS	10000	-828000.	780000.	2340.	$^{6380.}_{-1120.}$	2520.	$^{5370.}_{108.}$	210000.	$^{214000.}_{207000.}$	200000.	$^{206000.}_{193000.}$	1.41	$^{1.53}_{1.28}$	-0.0125	$^{0.0576}_{-0.0827}$
Mz_FUS	10000	-230000.	238000.	469.	$^{965.}_{-239.}$	26.8	$^{354.}_{-213.}$	40400.	$^{41400.}_{39500.}$	23000.	$^{23800.}_{22400.}$	5.22	$^{5.70}_{4.81}$	0.188	$^{0.297}_{0.0187}$

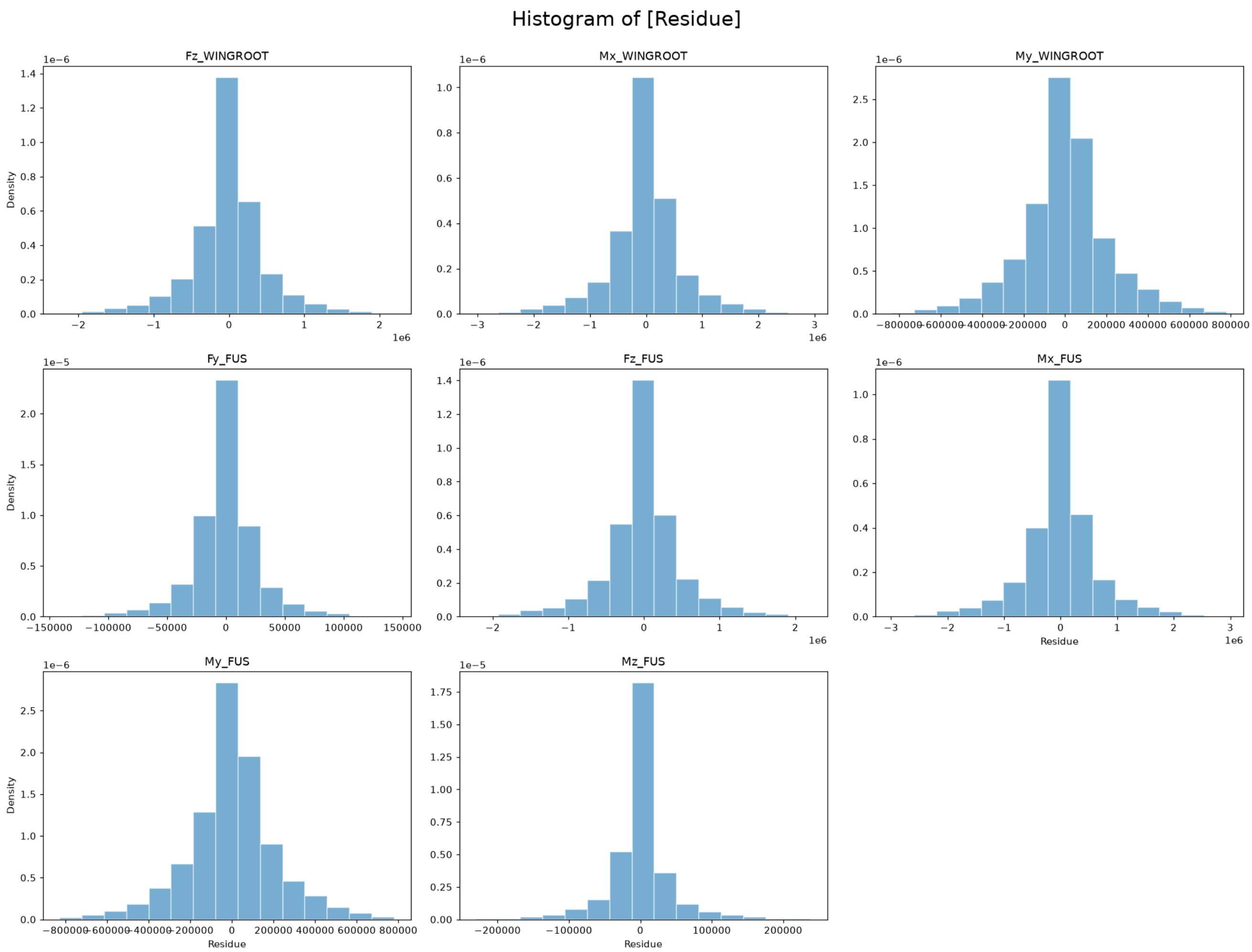
Residue statistics with bootstrap confidence intervals (superscript/subscript bounds).

Absolute Error Statistics Table

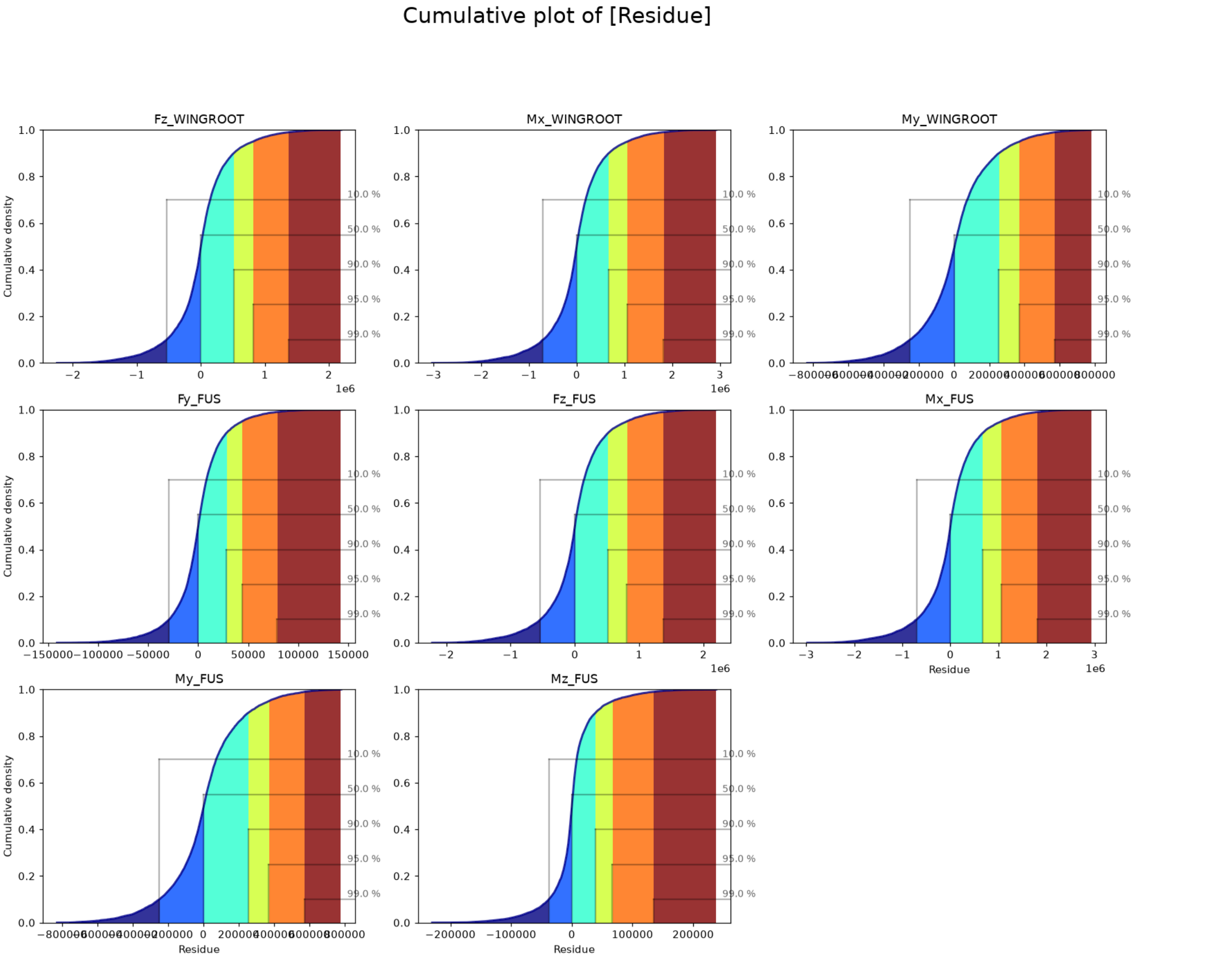
	count	min	max	mean	(BS)mean	median	(BS)median	std	(BS)std	IQR	(BS)IQR	kurtosis	(BS)kurtosis	skewness	(BS)skewness
Fz_WINGROOT	10000	42.6	2250000.	325000.	$^{331000.}_{319000.}$	198000.	$^{203000.}_{190000.}$	351000.	$^{358000.}_{344000.}$	362000.	$^{372000.}_{351000.}$	3.44	$^{3.76}_{3.09}$	1.82	$^{1.88}_{1.76}$
Mx_WINGROOT	10000	1.58	3030000.	424000.	$^{434000.}_{415000.}$	258000.	$^{264000.}_{248000.}$	463000.	$^{474000.}_{451000.}$	472000.	$^{484000.}_{452000.}$	3.53	$^{3.91}_{3.21}$	1.84	$^{1.90}_{1.79}$
My_WINGROOT	10000	26.1	834000.	150000.	$^{153000.}_{148000.}$	100000.	$^{103000.}_{97500.}$	147000.	$^{149000.}_{145000.}$	175000.	$^{181000.}_{172000.}$	1.98	$^{2.17}_{1.79}$	1.48	$^{1.52}_{1.43}$
Fy_FUS	10000	1.61	143000.	18100.	$^{18500.}_{17800.}$	11200.	$^{11500.}_{10900.}$	19800.	$^{20200.}_{19200.}$	19400.	$^{19900.}_{18700.}$	4.79	$^{5.26}_{4.20}$	2.02	$^{2.09}_{1.93}$
Fz_FUS	10000	12.0	2220000.	324000.	$^{330000.}_{316000.}$	197000.	$^{203000.}_{193000.}$	352000.	$^{360000.}_{343000.}$	364000.	$^{376000.}_{350000.}$	3.43	$^{3.73}_{3.12}$	1.82	$^{1.87}_{1.76}$
Mx_FUS	10000	26.5	2980000.	424000.	$^{432000.}_{415000.}$	256000.	$^{262000.}_{249000.}$	465000.	$^{473000.}_{450000.}$	476000.	$^{491000.}_{461000.}$	3.52	$^{3.94}_{3.12}$	1.84	$^{1.91}_{1.76}$
My_FUS	10000	12.6	828000.	150000.	$^{153000.}_{147000.}$	99800.	$^{104000.}_{96900.}$	147000.	$^{150000.}_{144000.}$	175000.	$^{180000.}_{170000.}$	1.98	$^{2.19}_{1.72}$	1.48	$^{1.53}_{1.42}$
Mz_FUS	10000	0.821	238000.	24400.	$^{24900.}_{23800.}$	11500.	$^{12000.}_{11200.}$	32200.	$^{33000.}_{31200.}$	26100.	$^{27100.}_{25000.}$	6.42	$^{7.08}_{5.76}$	2.37	$^{2.46}_{2.29}$

Absolute-error statistics with bootstrap confidence intervals.

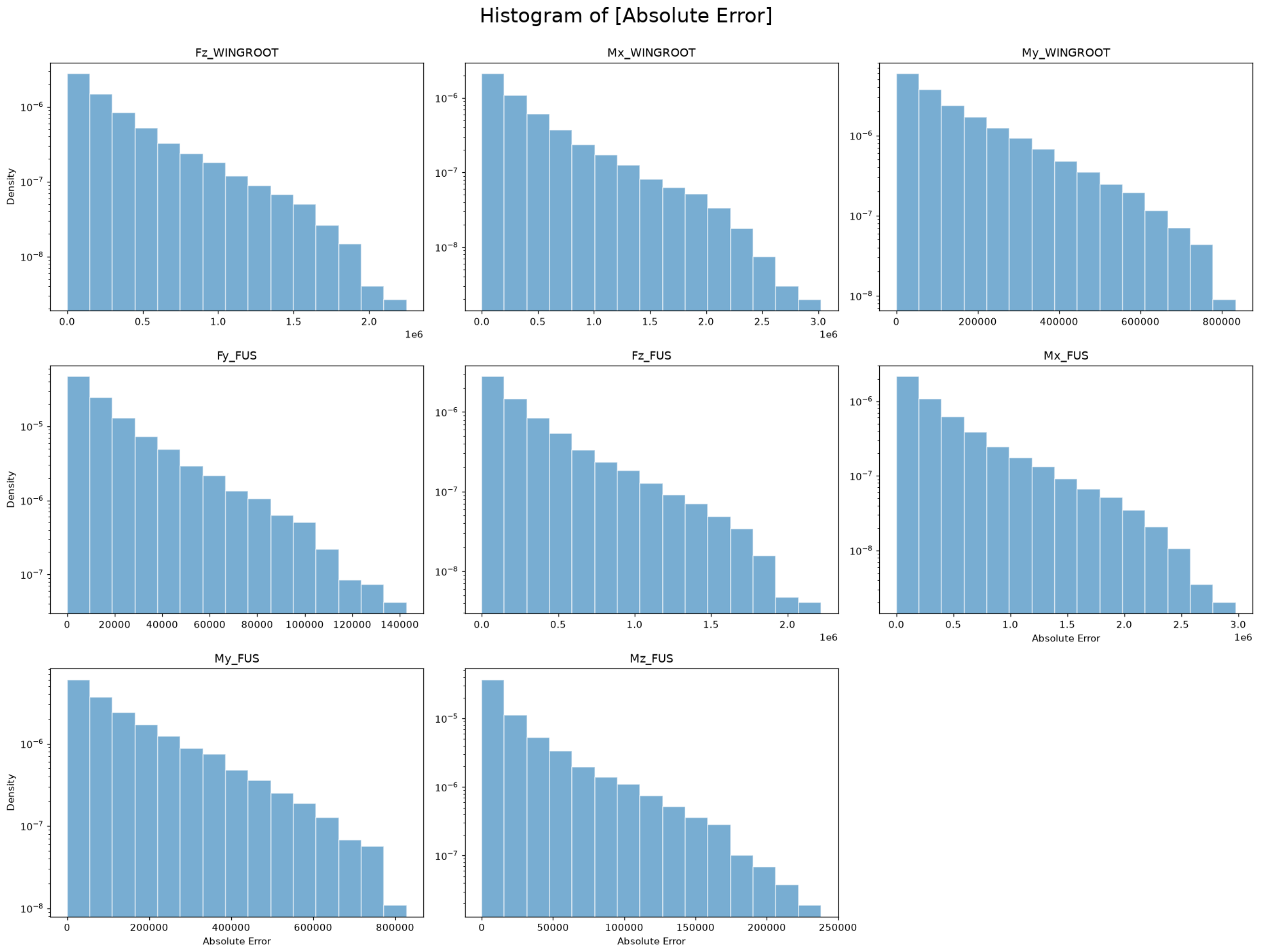
Residue Histograms



Residue CDF



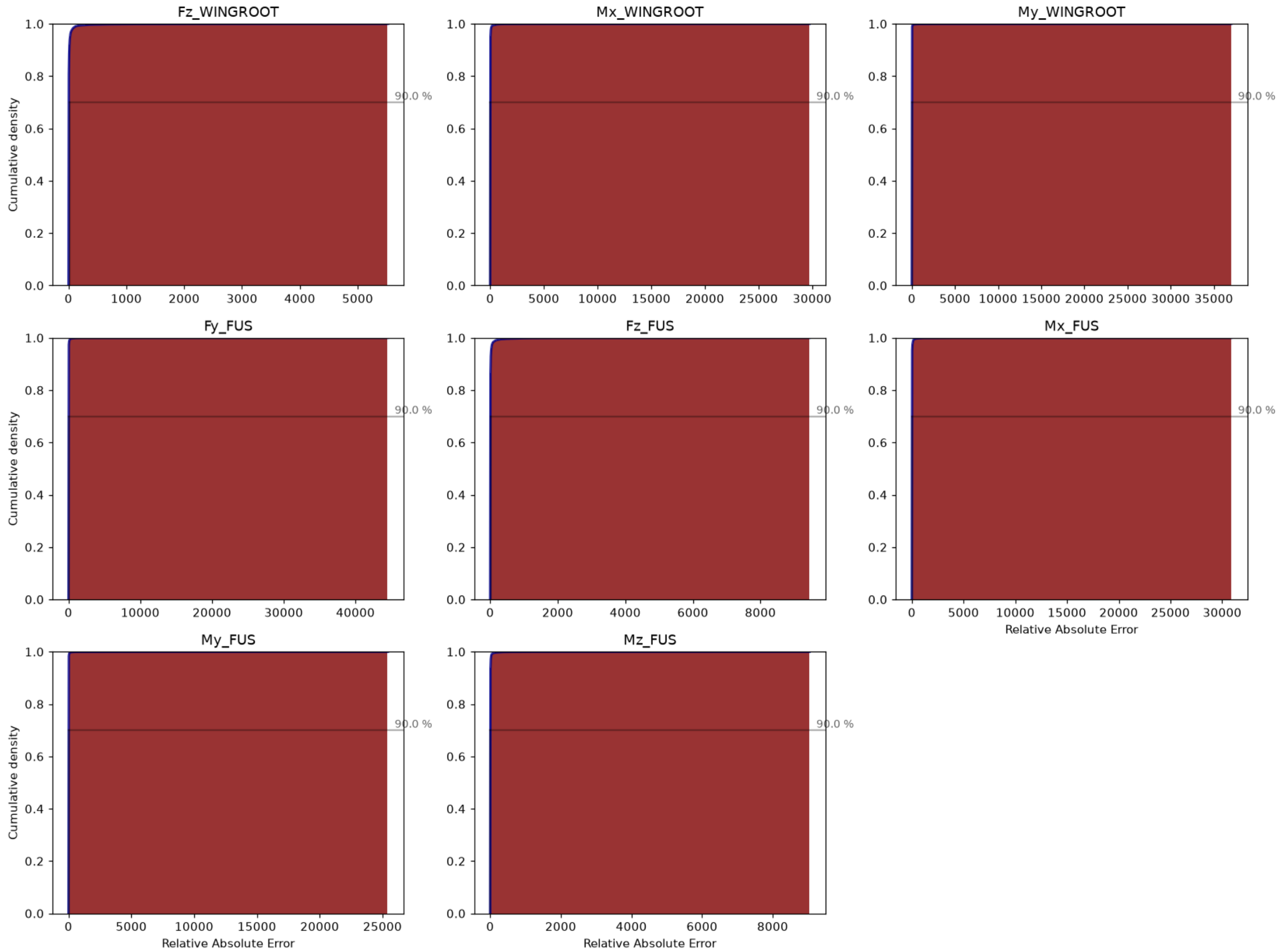
Absolute Error Histograms



Relative Absolute Error CDF — Q90 Accuracy Requirement

Orange dotted = observed Q90; red dashed = target threshold. Green title = PASS; red title = FAIL.

Cumulative plot of [Relative Absolute Error]

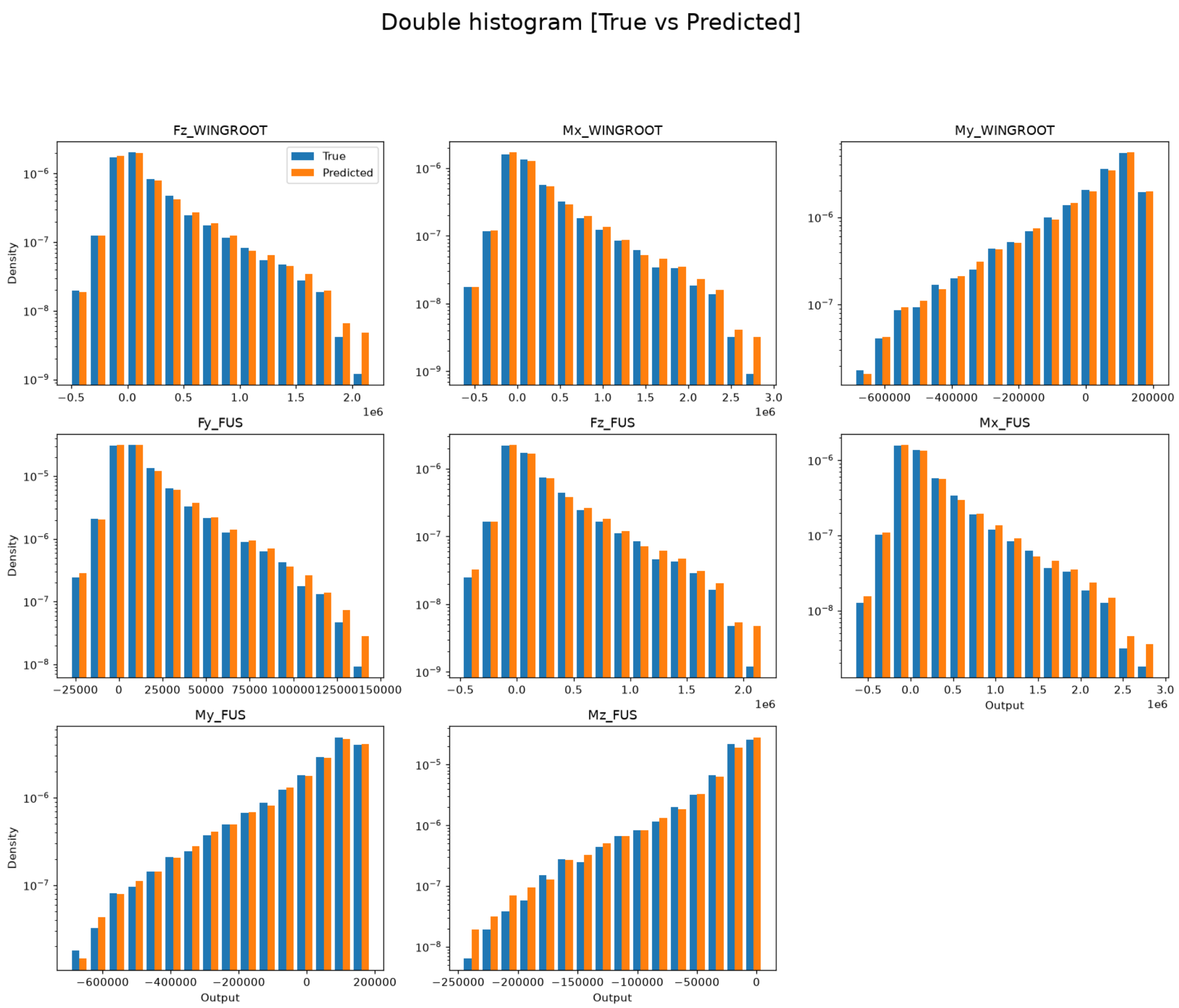


Output	Q90	Target	Status
Fz_WINGROOT	12.8856	10%	FAIL
Mx_WINGROOT	12.8081	10%	FAIL
My_WINGROOT	4.9689	10%	FAIL
Fy_FUS	10.8875	10%	FAIL
Fz_FUS	13.1123	10%	FAIL
Mx_FUS	13.3095	10%	FAIL
My_FUS	4.8529	10%	FAIL
Mz_FUS	5.7248	10%	FAIL

$p \geq 0.05$ — pass $p < 0.05$ — fail

True vs Predicted Distribution Comparison

AD and KS tests comparing the distribution of true vs. predicted values on the test set. A significant difference indicates the model is not reproducing the output distribution faithfully.

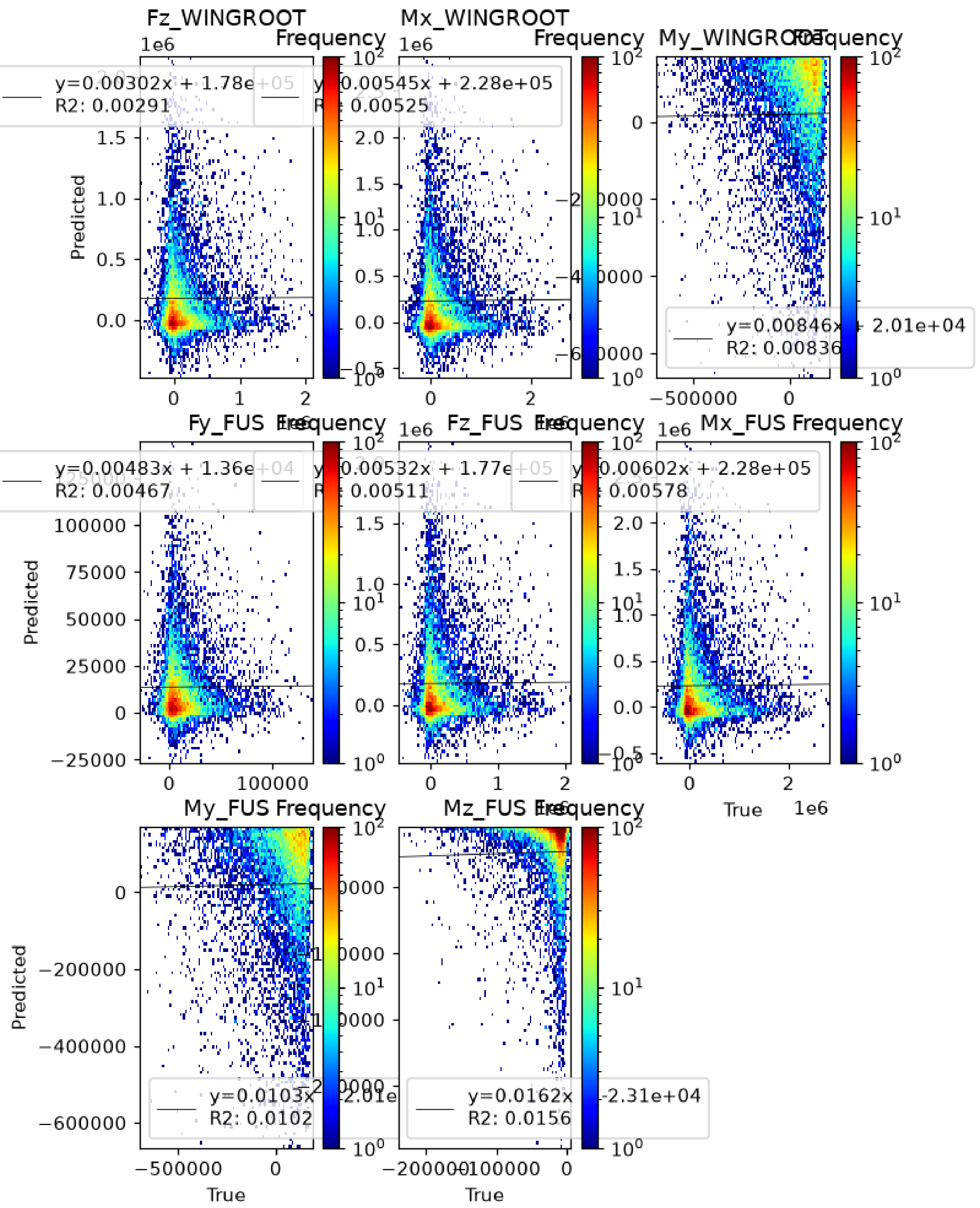


	AD	KS
Fz_WINGROOT	0.00425920151699012	0.0021998068154162414
Mx_WINGROOT	0.0011639430208941755	0.0005290265350566941
My_WINGROOT	0.001	7.133746468885948e-05
Fy_FUS	0.033711434511028565	0.03966362560959423
Fz_FUS	0.2309883074930256	0.2509732672225685
Mx_FUS	0.25	0.44629561636001003
My_FUS	0.035544925046655906	0.21692483428445863
Mz_FUS	0.001	1.329523599881888e-55

True vs predicted distribution tests. Red: $p < 0.05$.

True vs Predicted — 2D Histogram

Log-scaled frequency with a linear fit; the normalized slope should be close to 1.


Fz_WINGROOT
Mx_WINGROOT
My_WINGROOT
Frequency
y=0.00302x + 1.78e+05
R2: 0.00291
y=0.00545x + 2.28e+05
R2: 0.00525
y=0.00846x + 2.01e+04
R2: 0.00836
Fy_FUS
Fz_FUS
Mx_FUS
y=0.00483x + 1.36e+04
R2: 0.00467
y=0.00532x + 1.77e+05
R2: 0.00511
y=0.00602x + 2.28e+05
R2: 0.00578
My_FUS
Mz_FUS
y=0.0103x
R2: 0.0102
y=0.0162x
R2: 0.0156
Predicted
True
1e6

P(E|X) — Error Conditional on Inputs

We study whether the model error depends on the input variables. Two approaches are used: (1) a Kruskal-Wallis (KW) test on binned inputs (Sturges rule for bin count), and (2) violin plots showing the error distribution within each input bin.

Significant input-conditional bias ($p < 0.05$) detected for: Mach→Fz_WINGROOT, Mach→Mx_WINGROOT, Mach→My_WINGROOT, Mach→Fy_FUS, Mach→Fz_FUS, Mach→Mx_FUS, Mach→My_FUS, Mach→Mz_FUS, Altitude_ft→Fz_WINGROOT, Altitude_ft→Mx_WINGROOT.... These inputs explain residual variance — consider polynomial features or interaction terms.

Bias Detection Table (KW p-values)

Green ($p \geq 0.05$) = residue uniform across input bins; red ($p < 0.05$) = significant bias.

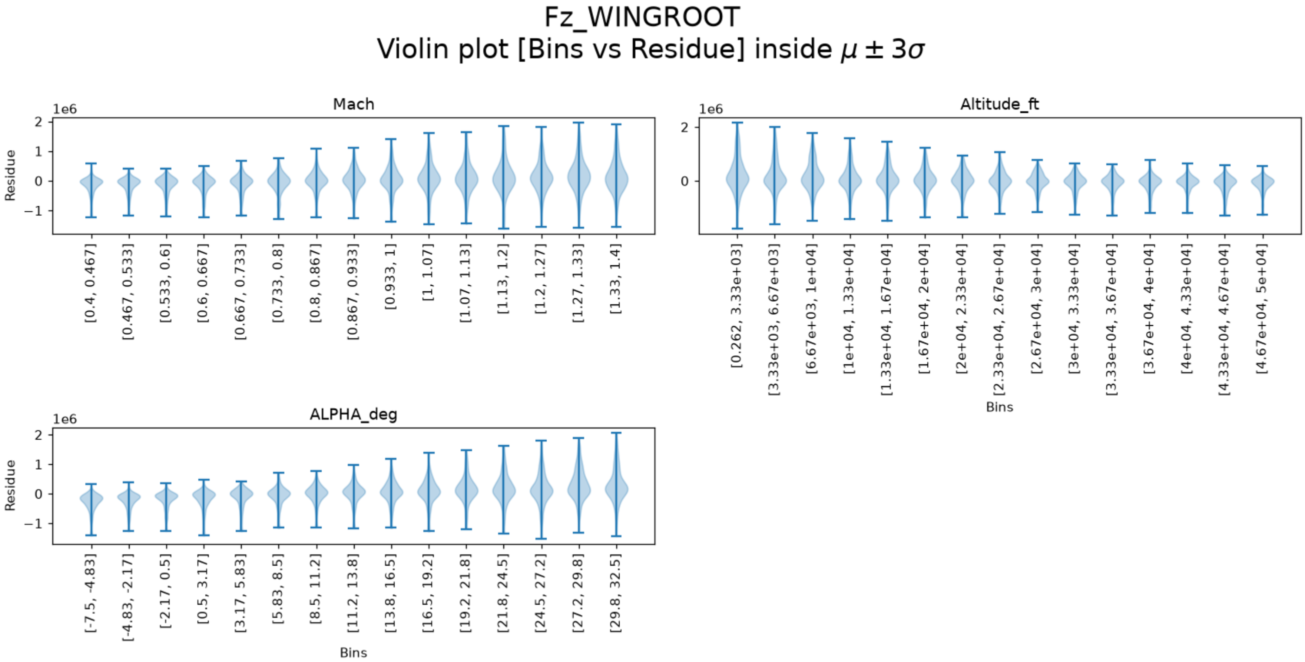
	Fz_WINGROOT	Mx_WINGROOT	My_WINGROOT	Fy_FUS	
Mach__binned	4.086310690277698e-156	1.825436181366889e-155	0.0	3.8463005238896635e-121	2.50635864
Altitude__ft__binned	8.460129592030745e-113	2.8863536299262887e-110	0.0	3.662162120065459e-120	7.38020180
ALPHA__deg__binned	0.0	0.0	0.015138540383883885	0.0	

Kruskal-Wallis p-values on Sturges-binned inputs. Red: $p < 0.05$ (error depends on that input).

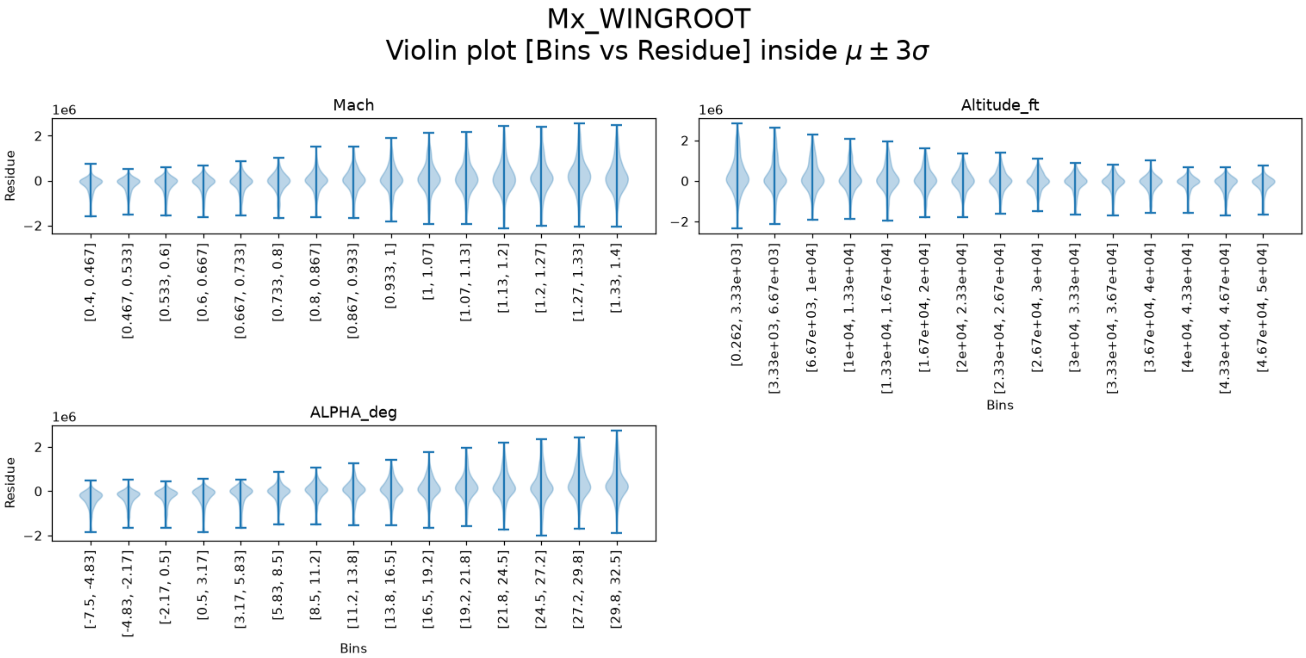
Residue Violin Plots per Output (binned by input)

Each panel shows the residue distribution for each input bin. A flat median line (dashed at 0) and symmetric violins indicate absence of bias.

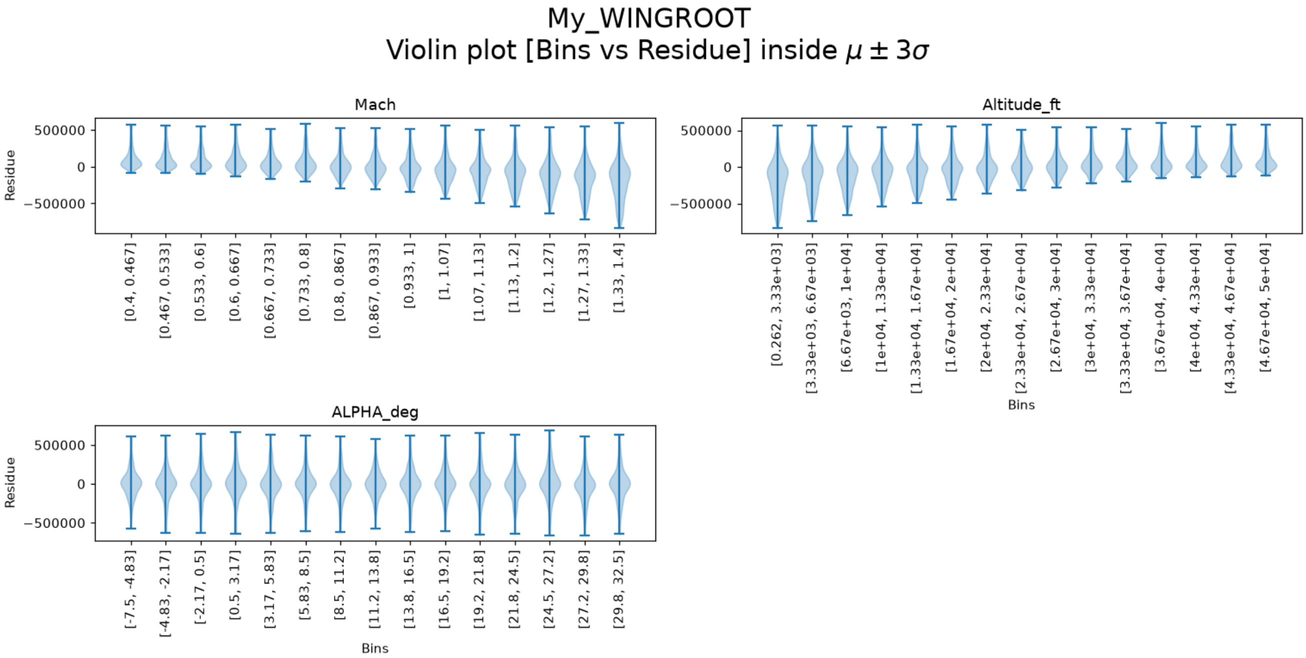
Residue — Fz_WINGROOT



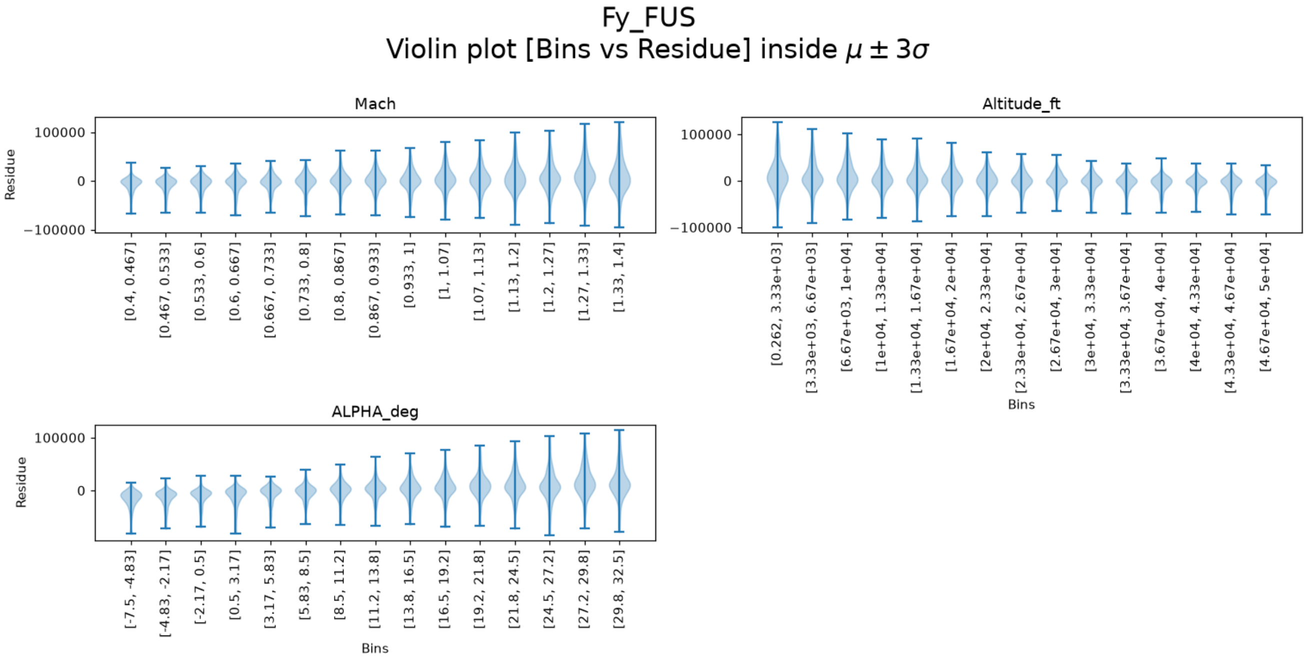
Residue — Mx_WINGROOT



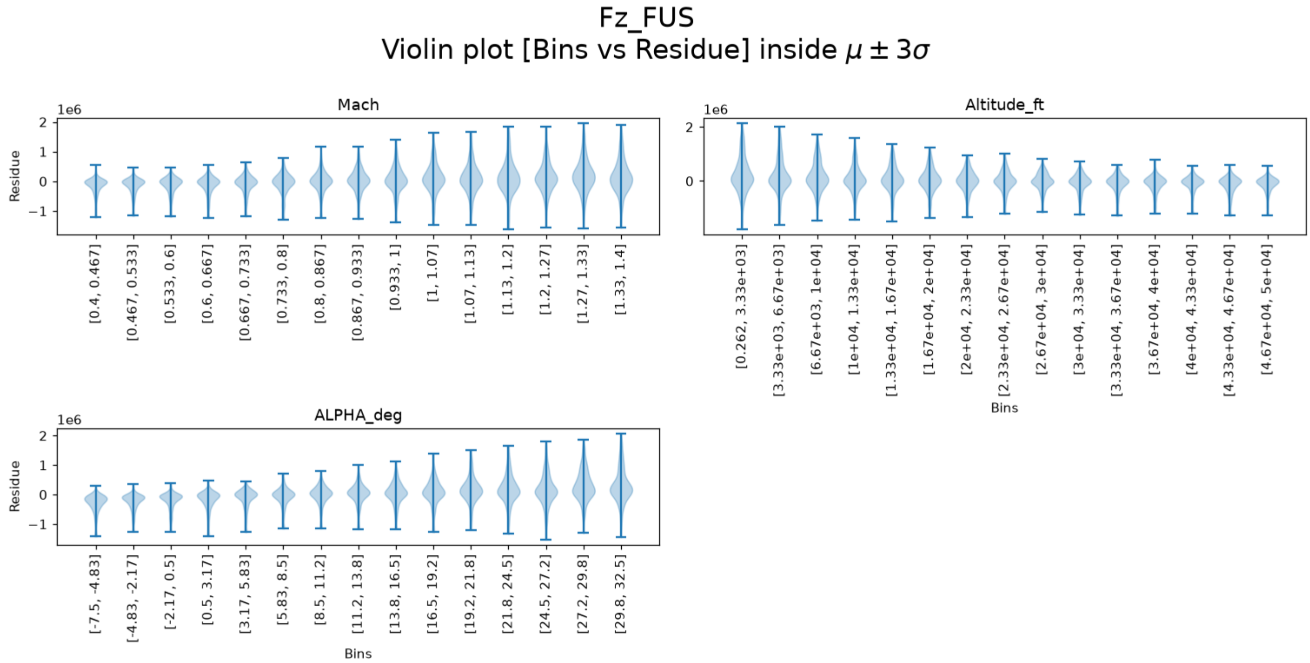
Residue — My_WINGROOT



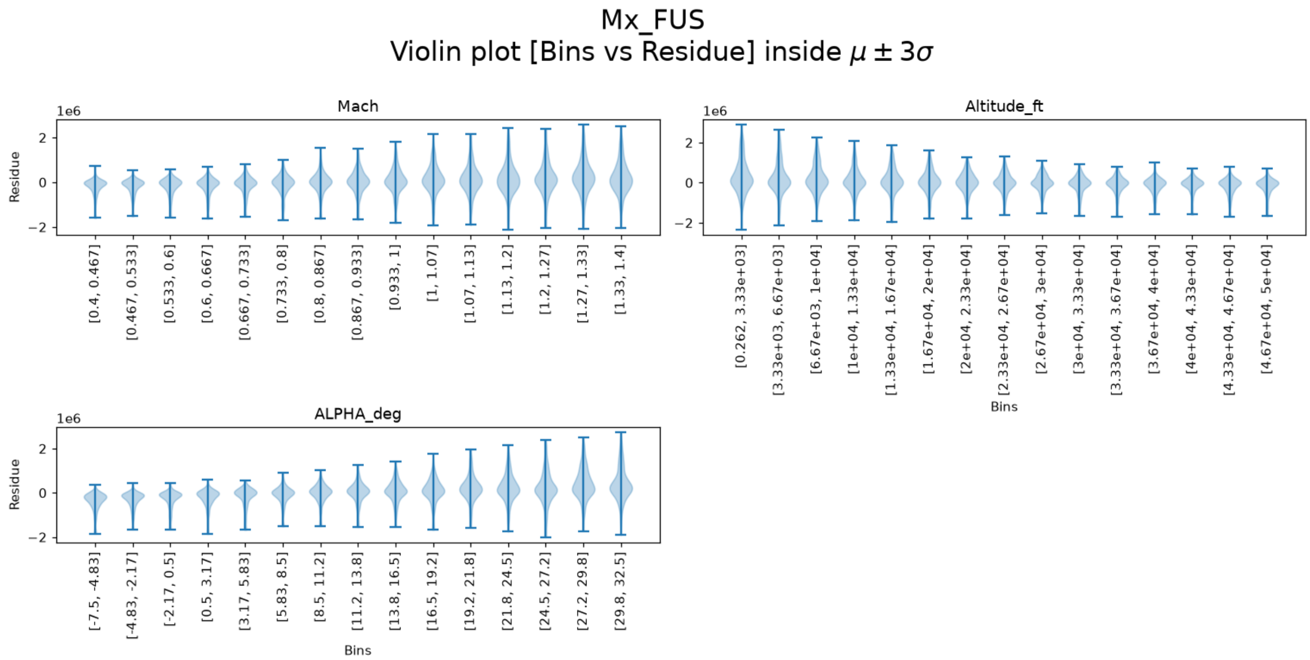
Residue — Fy_FUS



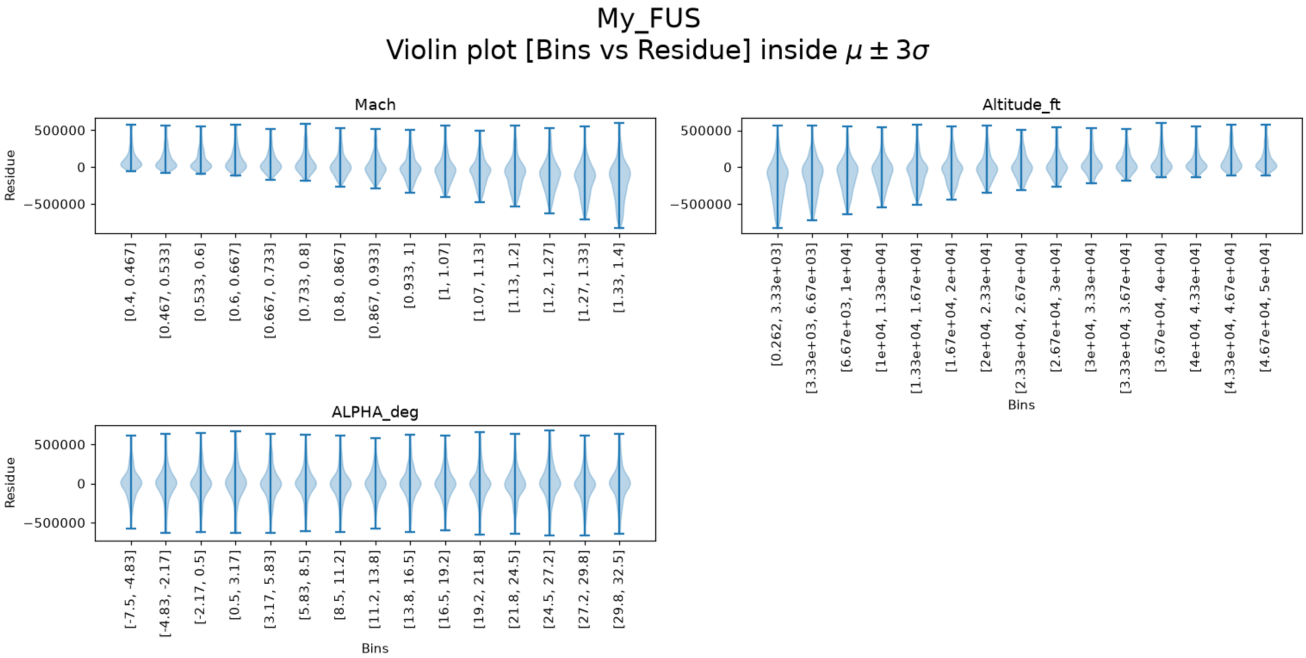
Residue — Fz_FUS



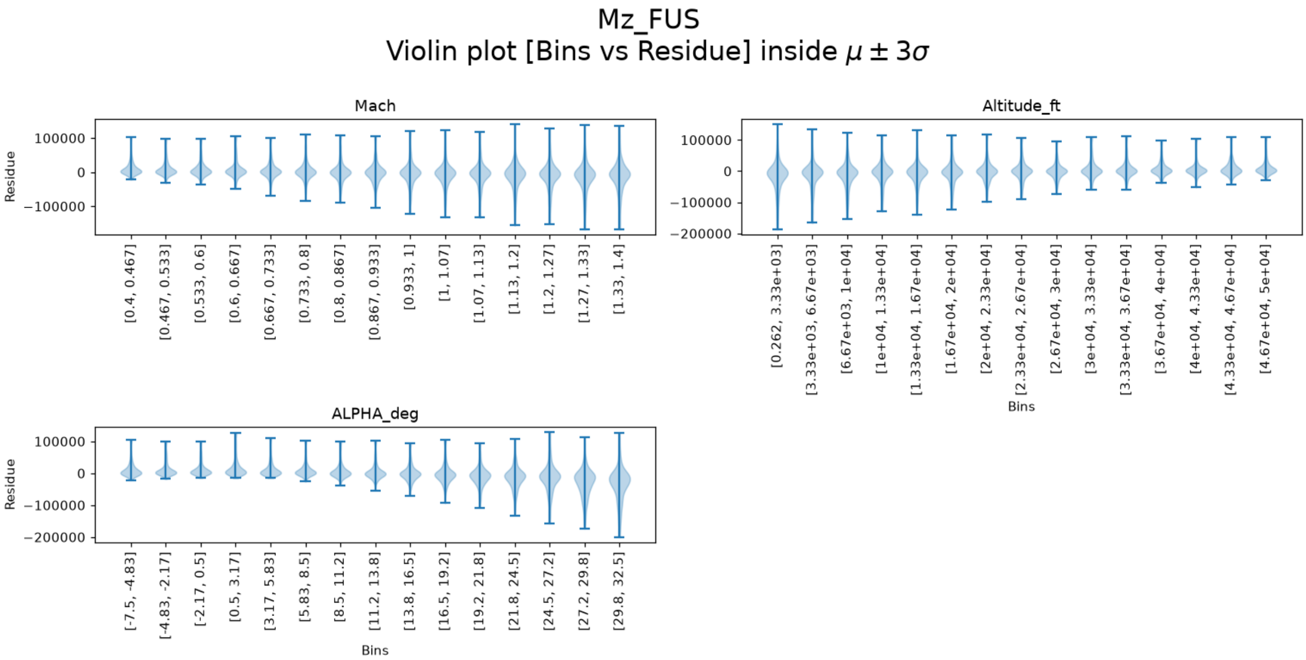
Residue — Mx_FUS



Residue — My_FUS

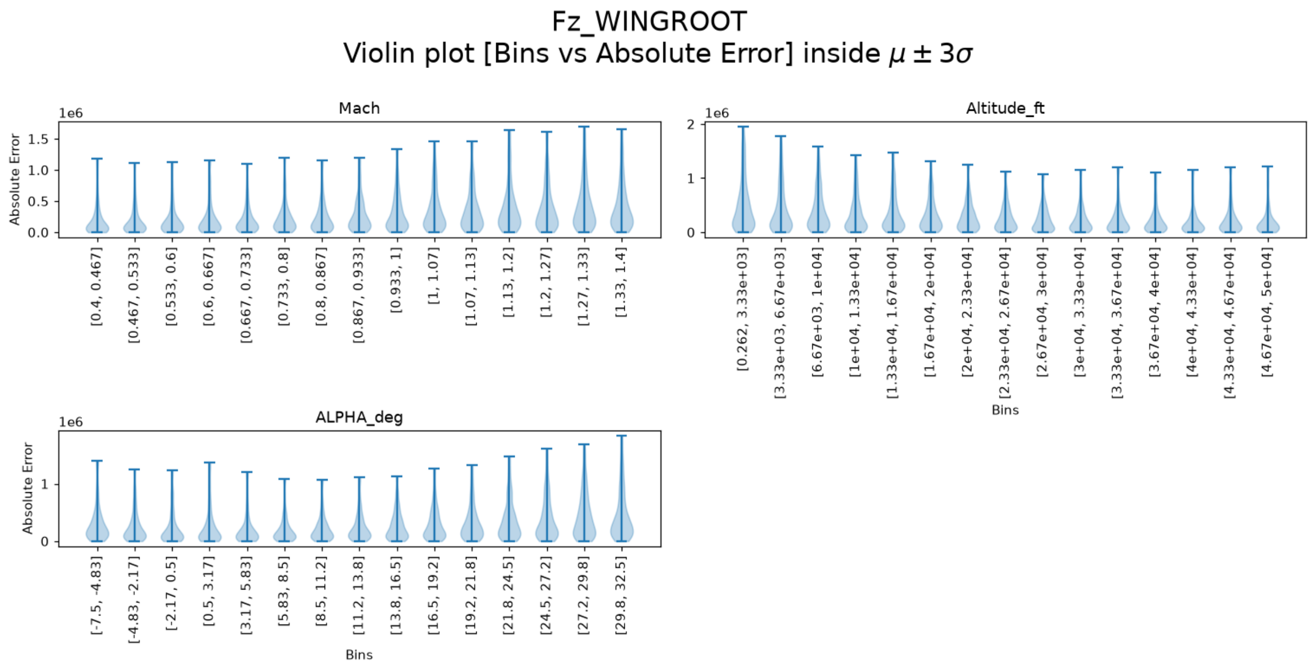


Residue — Mz_FUS

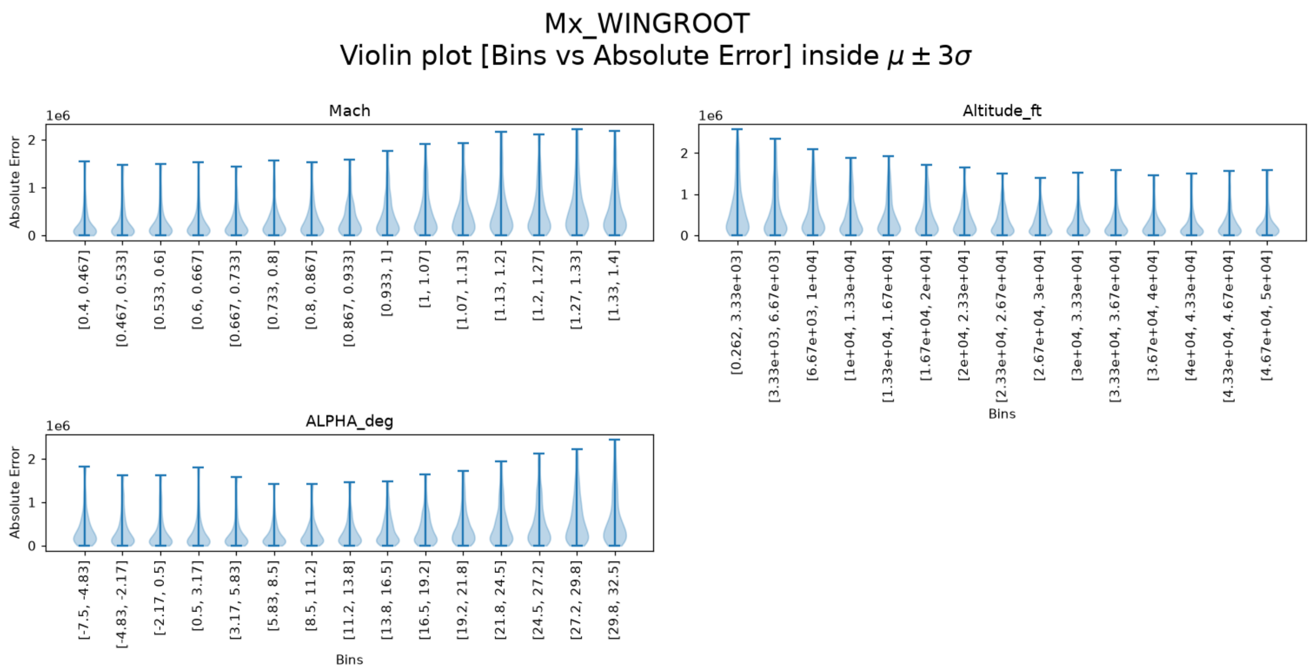


Absolute Error Violin Plots per Output (binned by input)

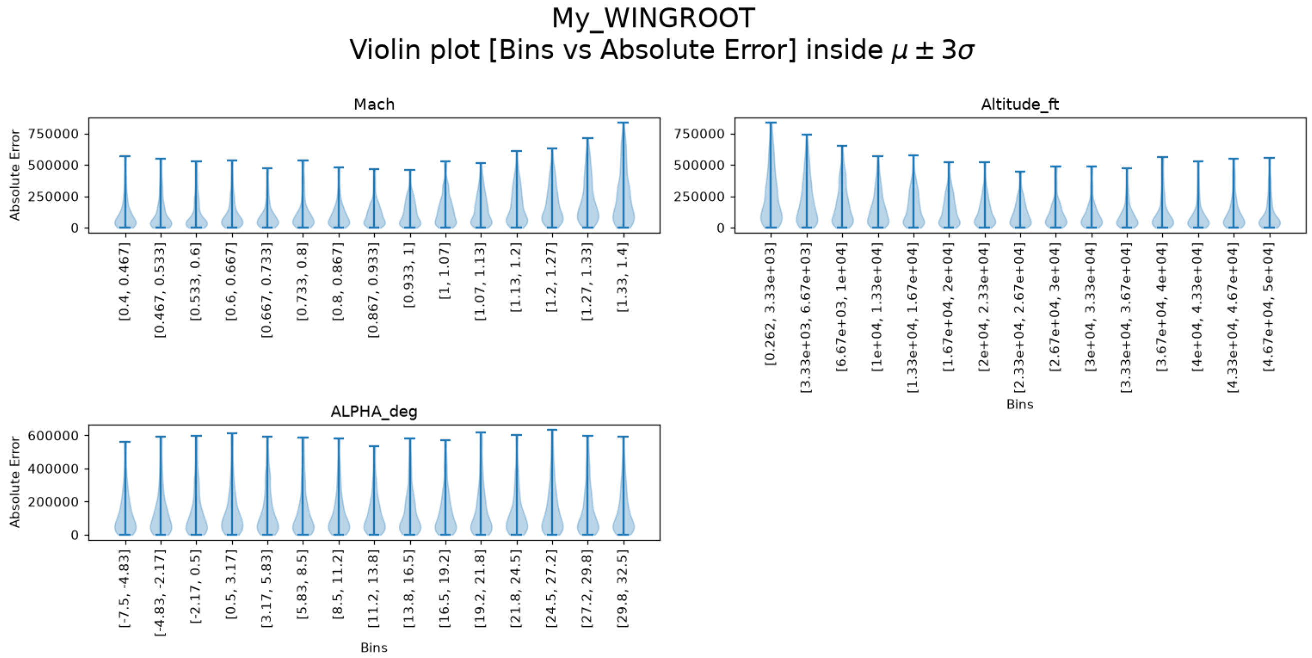
Absolute Error — Fz_WINGROOT



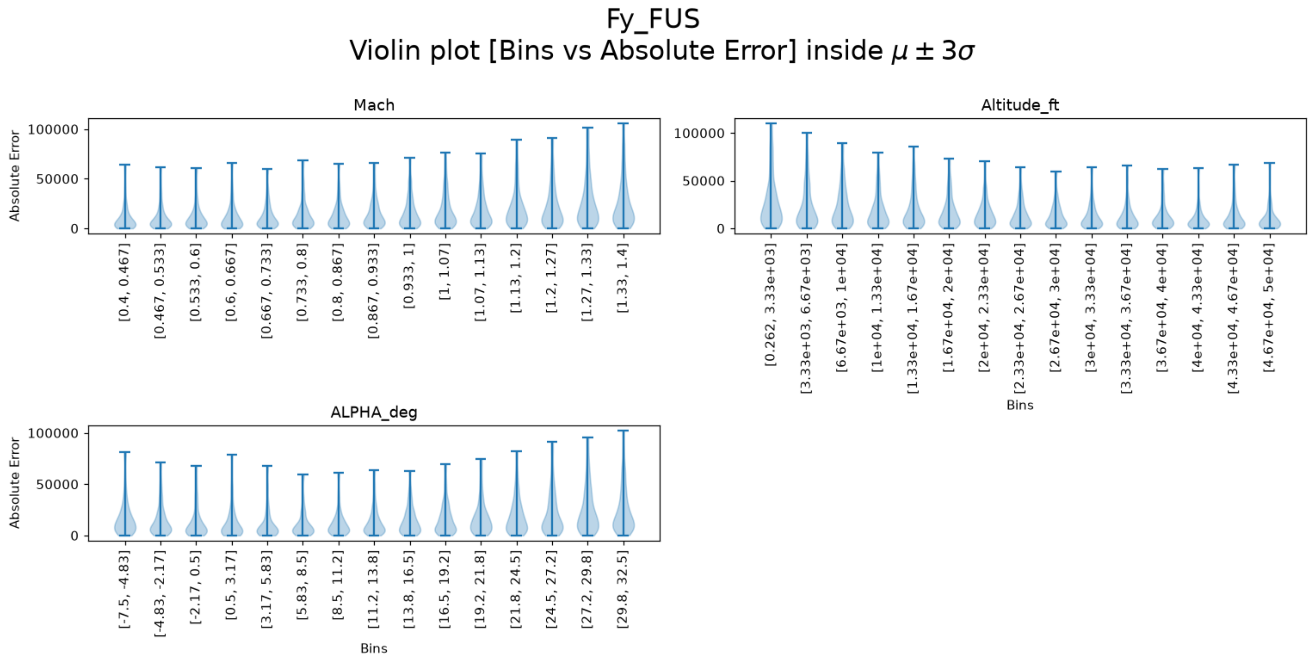
Absolute Error — Mx_WINGROOT



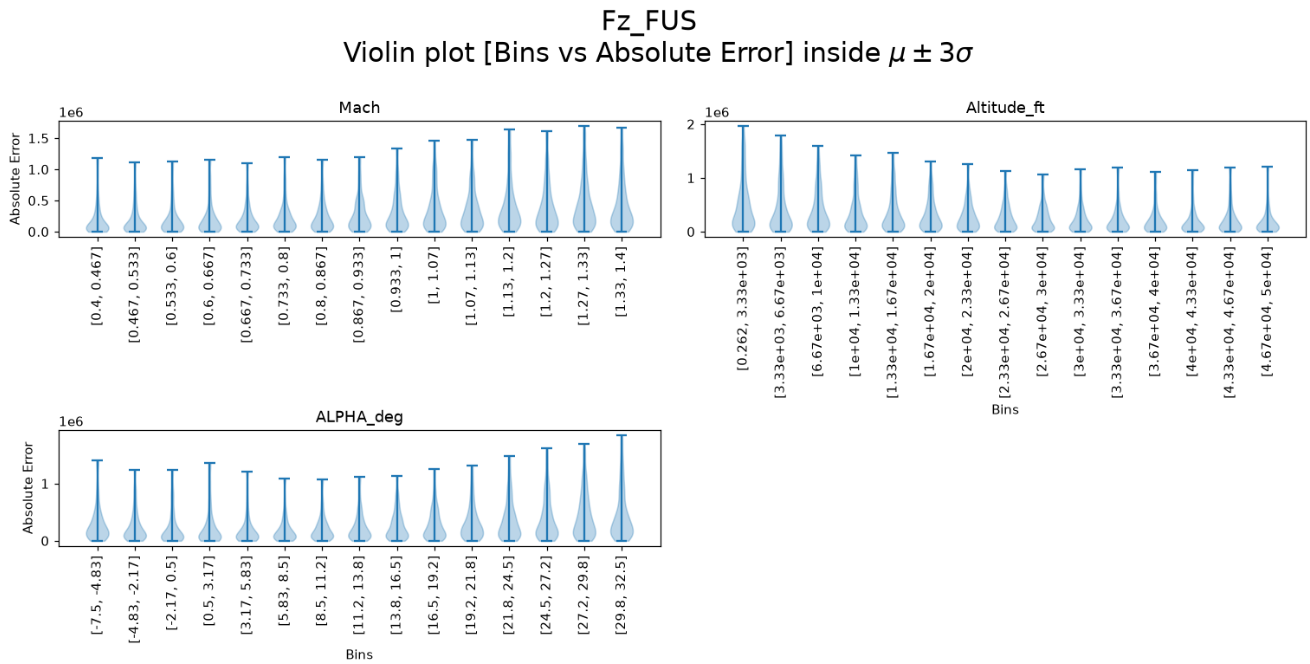
Absolute Error — My_WINGROOT



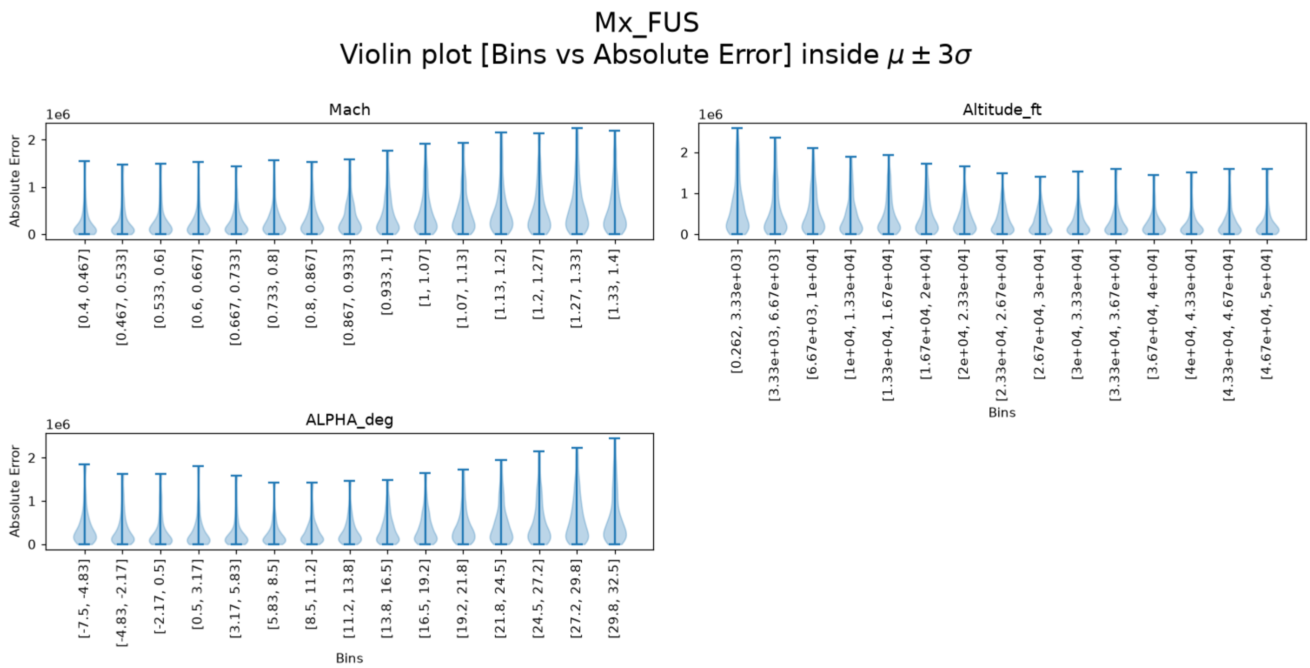
Absolute Error — Fy_FUS



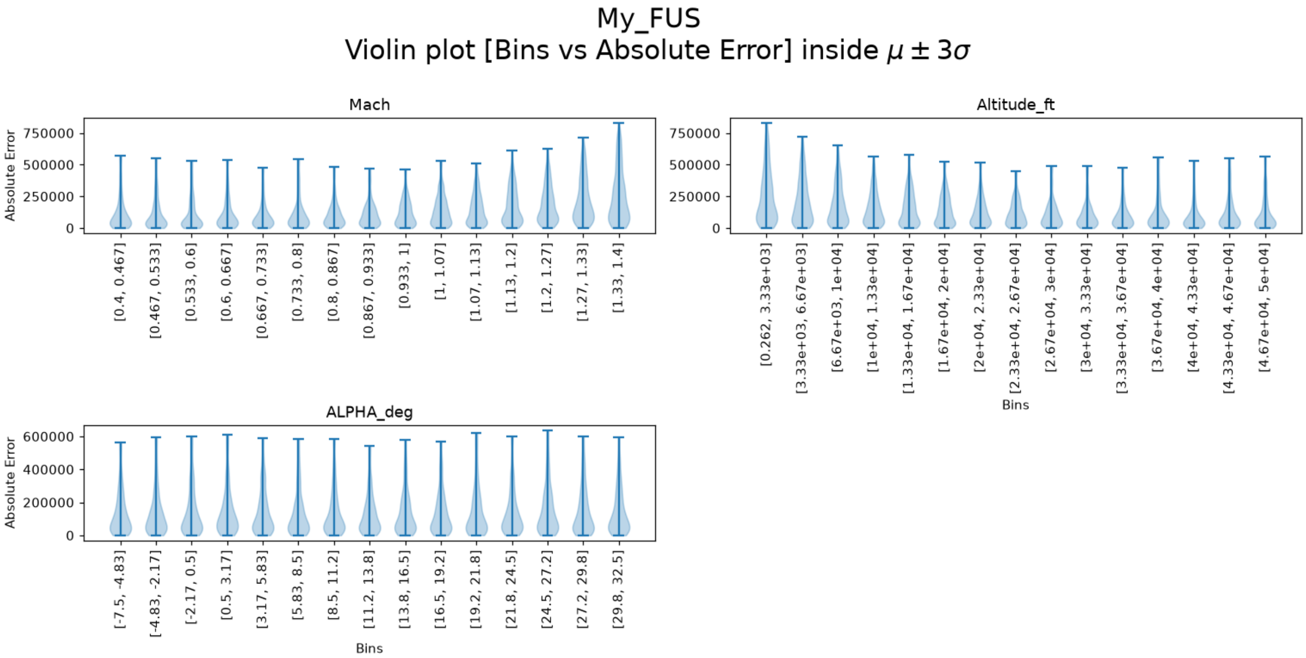
Absolute Error — Fz_FUS



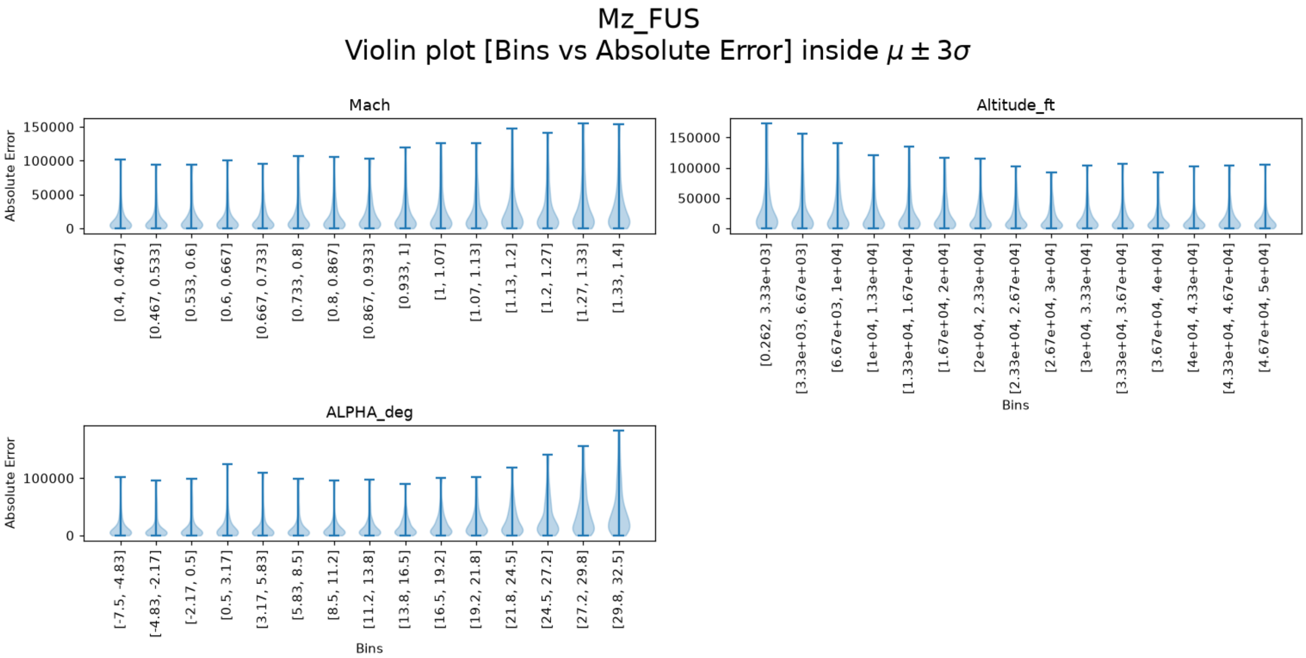
Absolute Error — Mx_FUS



Absolute Error — My_FUS



Absolute Error — Mz_FUS

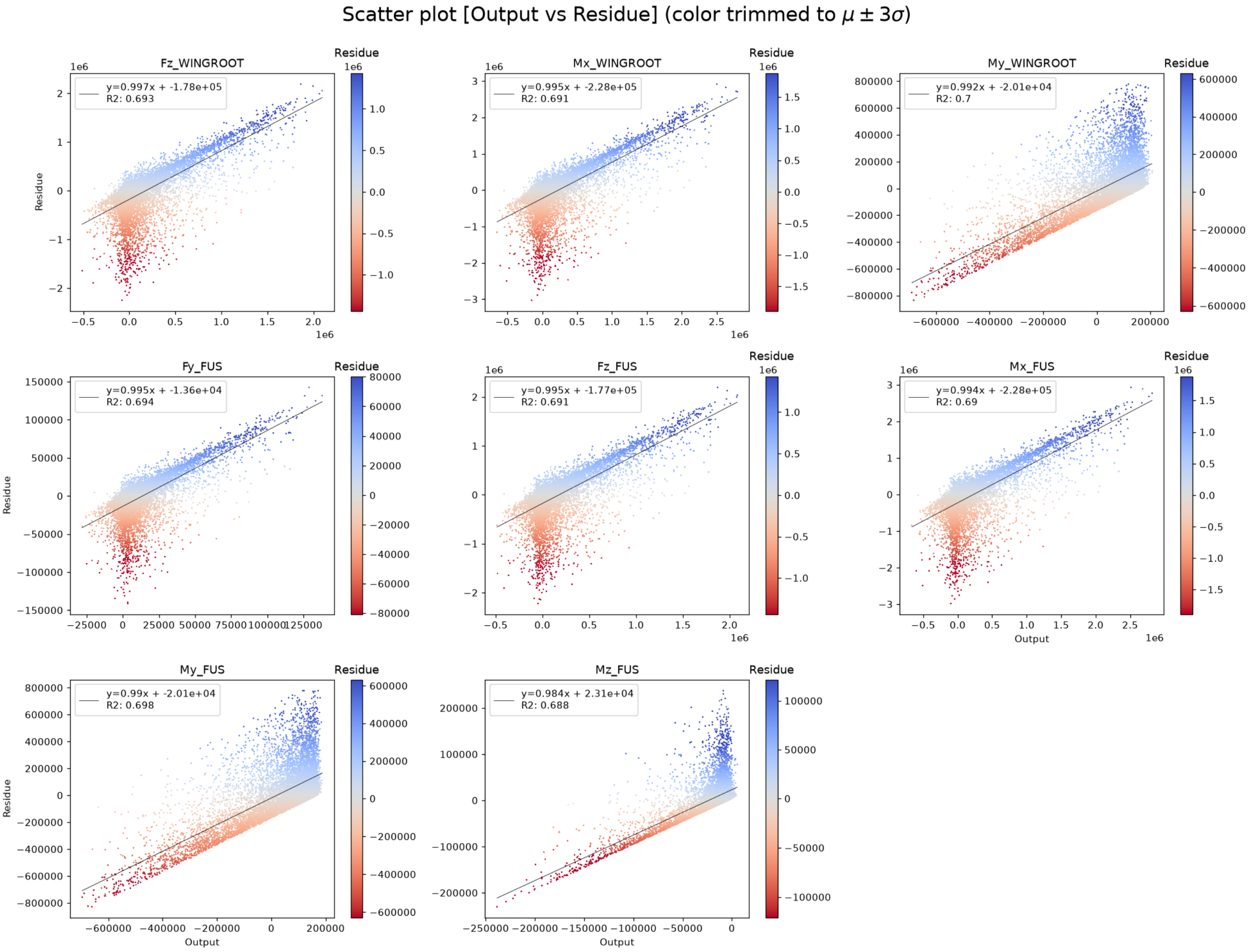


P(E|Y) — Error Conditional on Outputs

We study whether the error depends on the output magnitude. A significant trend (Pearson or Spearman $p < 0.05$) means the model is more accurate in some output ranges than others — a form of heteroscedasticity.

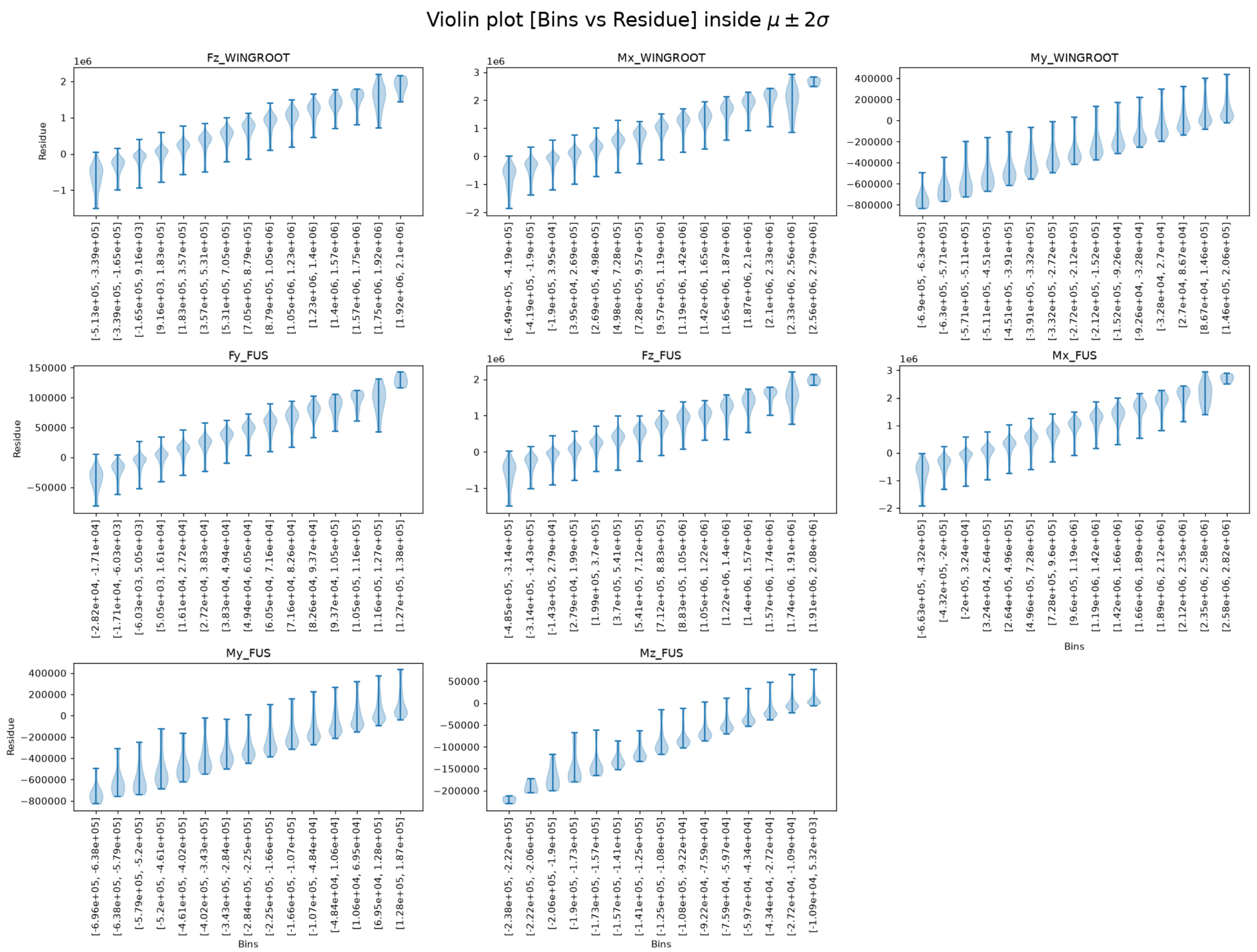
Residue vs True Output — Scatter

Pearson r and p-value shown in title. Red title = significant linear trend.



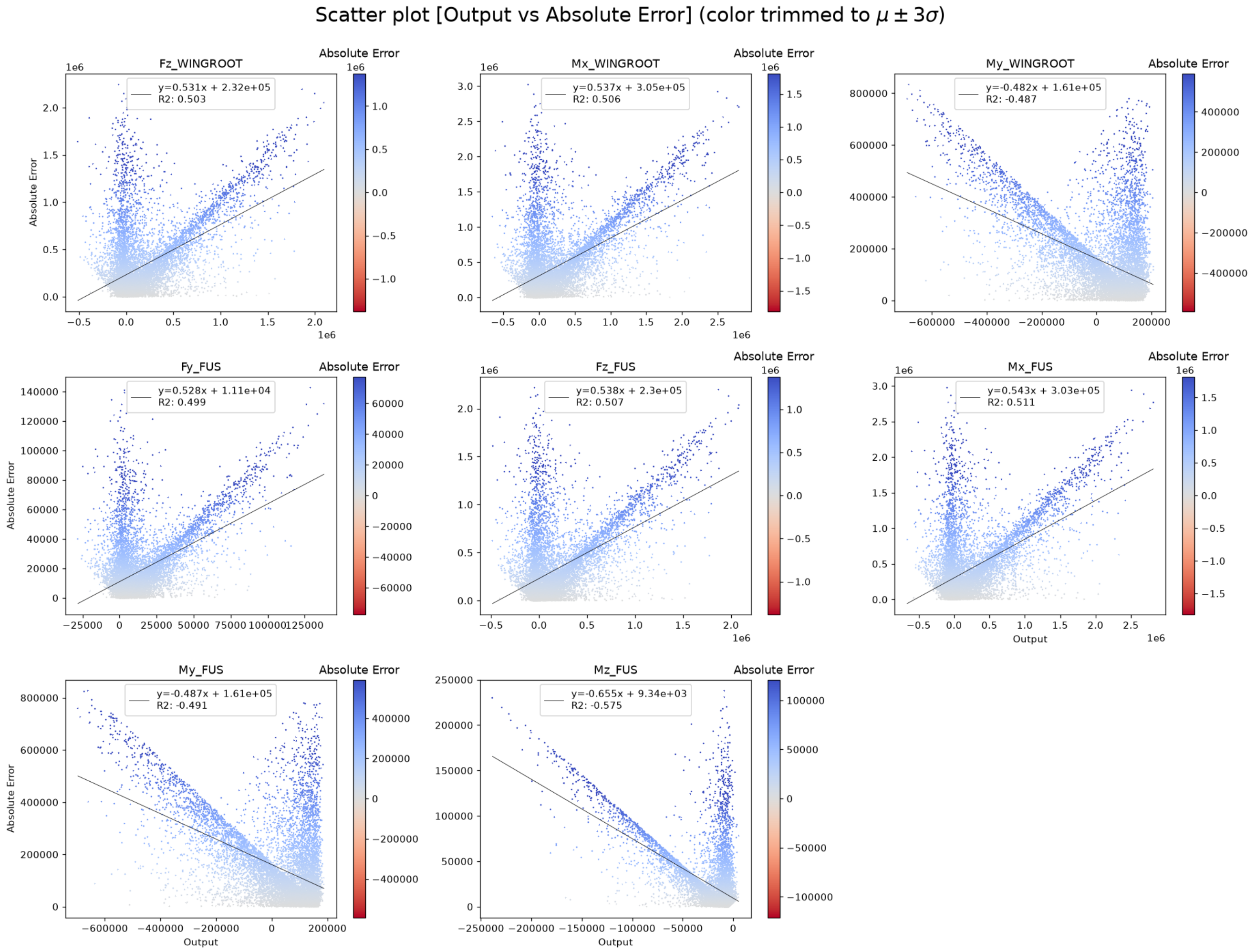
Residue Violin vs True Output Bins

Bins determined by Sturges rule. Median line dashed at 0.

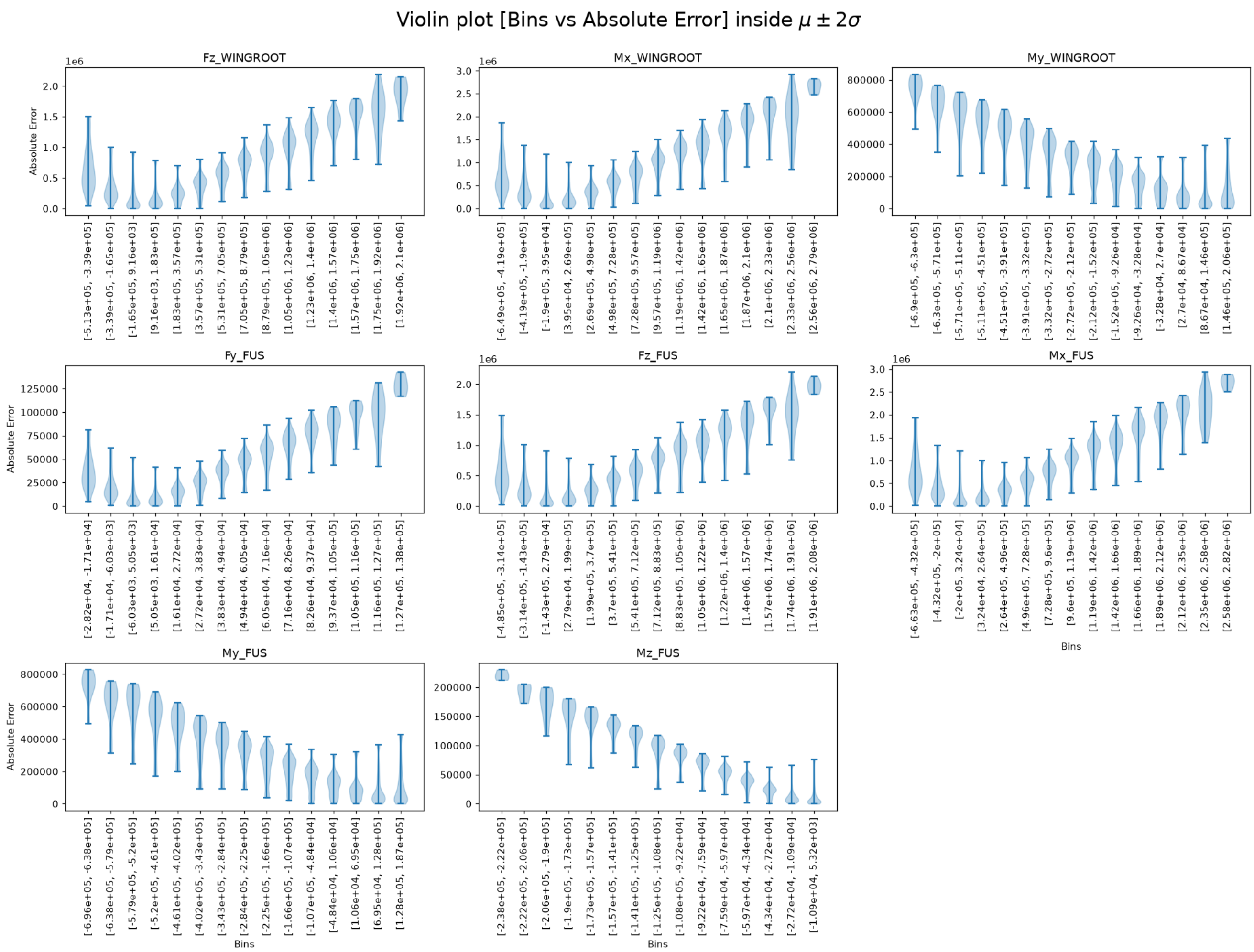


Absolute Error vs True Output — Scatter

Spearman r shown. Red title = significant monotonic trend.



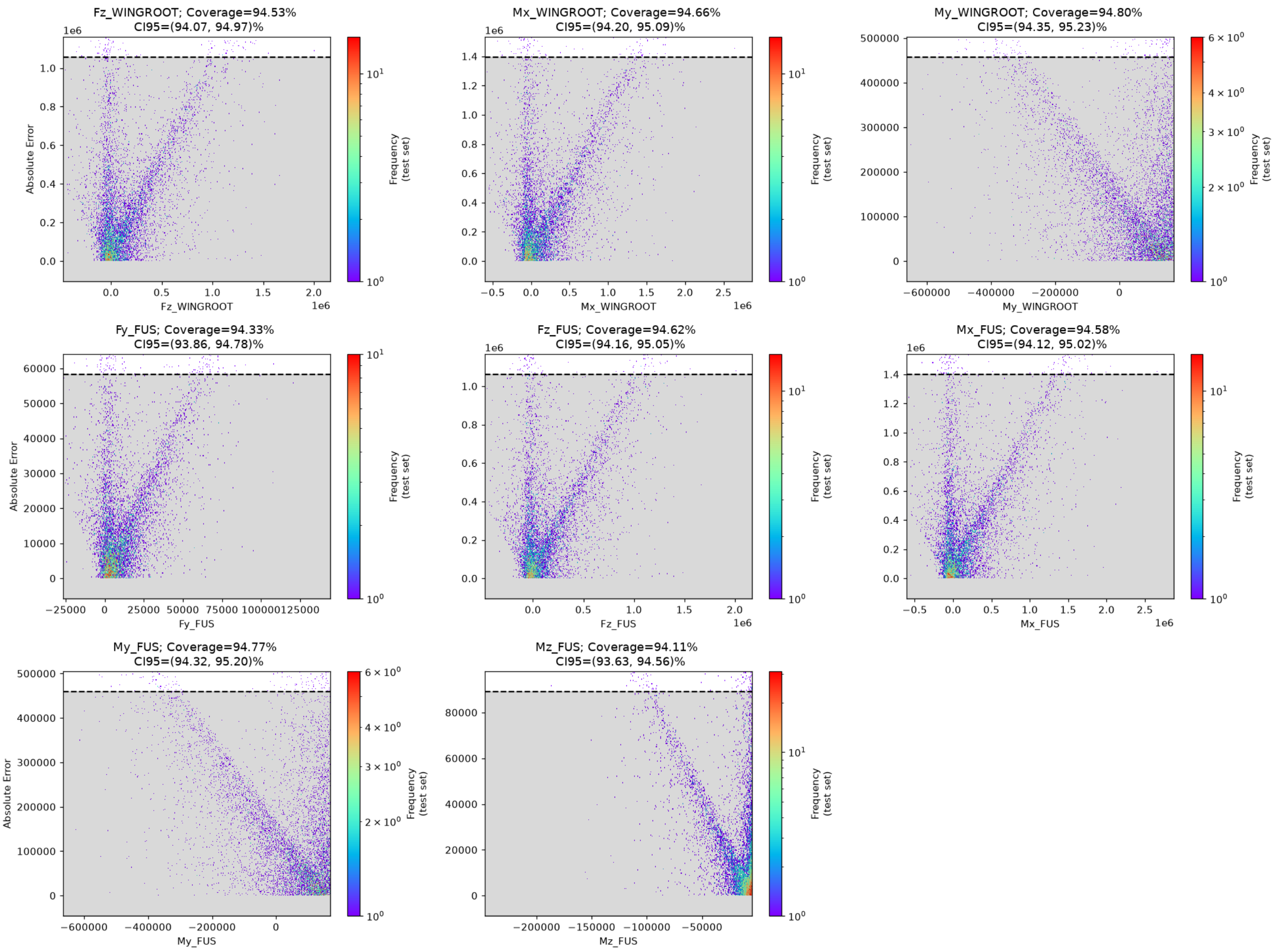
Absolute Error Violin vs True Output Bins


Violin plot [Bins vs Absolute Error] inside $\mu \pm 2\sigma$
Fz_WINGROOT
Mx_WINGROOT
My_WINGROOT
Fy_FUS
Fz_FUS
Mx_FUS
My_FUS
Mz_FUS
Absolute Error
Bins

Uncertainty Analysis

A global binned uncertainty model (95th percentile, right-tailed) is trained on the validation split of the absolute error and evaluated on the held-out test split, as in the extended validation notebook. Coverage is the proportion of test points whose error falls inside the predicted bound; it should be close to 95%.

Coverage Plot



Coverage Table

	Fz_WINGROOT	Mx_WINGROOT	My_WINGROOT	Fy_FUS	Fz_FUS	Mx_FUS	My_FUS	Mz_FUS	TOTAL COV
Coverage	94.53	94.66	94.8	94.33	94.62	94.58	94.77	94.11	nan
CI	[94.07, 94.97]	[94.2, 95.09]	[94.35, 95.23]	[93.86, 94.78]	[94.16, 95.05]	[94.12, 95.02]	[94.32, 95.2]	[93.63, 94.56]	nan

Empirical coverage of the 95th-percentile uncertainty bound on the test split.